

Chapter1

TITLE

**Development of a Web and Mobile Employment System with Smart Matching,
Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled
Workers and Households in Tayabas City**

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INTRODUCTION

The employment of domestic helpers and skilled workers in Tayabas City, Quezon has long depended on informal arrangements such as personal referrals and word-of-mouth recommendations, leaving both households and workers without a reliable, transparent, and organized means of connecting with one another. According to the Philippine Institute for Development Studies (PIDS, 2022), digital labor platforms remain limited in many semi-urban communities in the Philippines, resulting in restricted access to organized employment opportunities and reliable hiring systems. This absence of a structured local employment platform has contributed to persistent challenges including information asymmetry, lack of worker verification, and limited access to job opportunities. To address these gaps, the researchers propose the development of a web and mobile employment system that will centralize and digitize the hiring process for households, domestic helpers, and skilled workers in Tayabas City, incorporating smart matching, proximity-aware job allocation, and reputation scoring to ensure an inclusive, efficient, and trustworthy employment ecosystem for the community.

Project Context

Digital labor platforms have significantly transformed the labor market, particularly in the way workers and employers connect. According to the International Labour Organization (ILO, 2021), the number of digital labor platforms worldwide has increased fivefold over the past decade. In developing countries such as the Philippines, these platforms are considered a promising source of employment opportunities, enabling workers — including women, young people, and persons with disabilities — to access income-generating work. Despite this global trend, localized employment platforms that address the specific socioeconomic conditions of municipalities and cities in the Philippines remain limited, particularly in

semi-urban areas such as Tayabas City.

One of the key issues motivating this study is the continued reliance on informal hiring practices within local communities in the Philippines. Soriano (2021) noted that the Philippine labor market is largely characterized by informal employment arrangements where workers and employers depend on personal referrals, social networks, and word-of-mouth to find opportunities. This situation is common in community-based services such as domestic work and household skilled labor, where formal recruitment systems are limited. Although these practices are familiar in many communities, they often result in a lack of standardized compensation, worker verification, and job security. The proposed system aims to address this gap by providing a structured and transparent digital platform for connecting households and workers in Tayabas.

Trust and accountability are also critical factors in online labor markets. A study by Hesse and Teubner (2023), published in *Electronic Markets*, examined how reputation signals such as ratings and reviews from completed transactions influence user trust and hiring decisions on gig economy platforms. The study found that workers with higher aggregated ratings are more frequently hired and are generally perceived as more reliable. These findings support the inclusion of a reputation scoring system in the proposed platform, allowing households to rate workers after job completion while also enabling workers to rate households in order to promote fairness and accountability on both sides.

The accessibility of digital services across devices has also been identified as a key factor in the adoption of employment platforms. The development of a mobile application component alongside the web platform ensures that users can access the system conveniently through their smartphones, which are among the most widely used devices in the Philippines. According to the Philippine Statistics Authority (PSA, 2023), mobile phone ownership continues to rise even in semi-urban and rural areas, making mobile applications

a practical and inclusive channel for delivering digital employment services to households and workers in Tayabas City.

This study aims to address these gaps by developing a centralized web and mobile employment system specifically designed for domestic helpers, skilled workers, and households in the City of Tayabas, Quezon. By integrating smart matching, proximity-aware job allocation, reputation scoring, and notification features, the proposed system seeks to modernize local employment practices, improve transparency, and create a more efficient and equitable employment system for the Tayabas community.

Statement of Objectives

The study aims to develop a web and mobile employment system with smart matching, proximity-aware job allocation, and reputation scoring for workers and households in Tayabas City, adhering to ISO/IEC 25010 Software Quality Standards (functionality, usability, reliability, performance, security, and maintainability).

Specifically, the study will aim to:

1. Identify the employment challenges and concerns experienced by households and workers in Tayabas in relation to:
 - 1.1. Availability and accessibility of qualified domestic helpers and skilled workers;
 - 1.2. Transparency in service rates and worker qualifications; and
 - 1.3. Communication and notification between households and workers.
2. Develop a web and mobile employment system that will address the identified challenges in terms of:
 - 2.1. Job posting and worker application management;
 - 2.2. Smart worker matching based on skills, availability, verification status, and geographic distance; and
 - 2.3. Profile management with mandatory service rate disclosure.

3. Design and implementation of Reputation and Trust Management System that will include:
 - 3.1. Mutual feedback between workers and households;
 - 3.2. Automated feedback reminders after job completion; and
 - 3.3. Neutral default rating assignment for unsubmitted feedback within a specified period.
4. Develop a Communication and Notification System that will ensure timely updates through:
 - 4.1. Email notifications for detailed account and job status updates; and
 - 4.2. In-system notifications within the platform dashboard.
5. Provide an Employment Analytics and Monitoring Dashboard that will display:
 - 5.1. Employment activity including job postings, active applications, and completed services;
 - 5.2. Geographic service distribution across barangays in Tayabas City; and
 - 5.3. Service rate transparency summaries per job category.
6. Evaluate the proposed system based on the ISO/IEC 25010 Software Quality Standard in terms of:
 - 6.1. Functional Suitability;
 - 6.2. Performance Efficiency;
 - 6.3. Interaction Capability (Usability);
 - 6.4. Reliability;
 - 6.5. Security;
 - 6.6 . Maintainability; and
 - 6.7 . Compatibility.

Scope and Limitations of the Study

This study focuses on the development of a web and mobile employment system designed to connect households with domestic helpers and skilled workers within Tayabas City, Quezon. The system aims to provide an organized and accessible platform where households can easily find workers while workers can access available employment opportunities. The intended users of the system include households, domestic helpers, skilled workers, and system administrators who are responsible for monitoring and managing the platform. The development and implementation of the system will be conducted within the duration of the research period.

The system will include several main components to support employment matching and platform management. These include a Household Dashboard for posting job requests and managing worker applications, a Worker Profile and Dashboard where workers can create accounts, present their skills, and apply for jobs, and a Job Listing and Application section where available job opportunities are displayed. An Administrative Dashboard will allow administrators to monitor system activities, verify user accounts, and manage services. The platform will also include an Employment Analytics Dashboard that presents summarized information such as job postings, completed services, verified users, and the distribution of services across barangays in Tayabas City. To improve efficiency and user experience, the system will include features such as Smart Matching and Worker Recommendation, which suggests workers based on skills, availability, verification status, and approximate distance from the job location. A Reputation and Review System will allow users to provide ratings and feedback after job completion. The system will also provide multi-channel notifications through email and in-system alerts. Additional features include a Distance Indicator that shows the approximate distance between the worker and the job site, service rate transparency in worker profiles, user verification monitoring, barangay employment distribution analytics, and Role-Based Access Control to manage system security. A mobile application component will be developed alongside the web platform to extend system accessibility to smartphone users, allowing households and workers to

manage their accounts, view job listings, and receive notifications conveniently through their mobile devices.

However, the system has several limitations. The platform will only be available to users within Tayabas City, Quezon. Most system features require internet connectivity, which may affect users with limited access to the internet. The distance indicator will only show approximate straight-line distance and will not calculate actual travel routes or time. The distance indicator will show the approximate driving distance between the worker's location and the job site using map routing services, though estimated travel time may vary depending on actual road conditions. User verification will depend on administrative review of submitted identification documents and will not be directly connected to government databases. In terms of payment handling, the system does not include built-in payment processing or online transactions. All payments will be arranged and settled directly between the household and the worker outside the platform. The system only serves as a medium for job matching and communication, and it does not monitor, record, or guarantee any financial transactions between users. In addition, the effectiveness of the platform relies on the active participation of users in applying for jobs and providing honest ratings and feedback. The service rates shown in worker profiles are only reference values, and the final hiring decision and pricing will still depend on the agreement between the household and the worker.

Purpose and Description

The proposed study, entitled "Development of a Web and Mobile Employment System with Smart Matching, Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled Workers and Households in Tayabas City," aims to develop a centralized, accessible, and efficient digital platform that will bridge the gap between domestic helpers and skilled workers and the households seeking their services within the City of Tayabas, Quezon. The primary purpose of the system is to modernize and organize the process of domestic and skilled worker employment by replacing informal and fragmented hiring methods with a structured and technology-driven solution. The platform is designed to cater to two primary user groups: households that require domestic or skilled services, and workers who offer such services. By bringing both parties together in a single

digital environment — accessible through both web browsers and mobile devices — the system aims to reduce hiring inefficiencies, increase transparency, and promote trust within the local employment ecosystem.

Once implemented, the proposed system is expected to provide significant improvements in employment coordination and service accessibility within the Tayabas community. The following groups are expected to benefit most from the system:

- **Household** – Households will benefit from a reliable and organized platform where they can post job requests, browse worker profiles, view service rates, and receive worker recommendations based on skills, availability, and proximity. The platform will eliminate the need for time-consuming and unreliable word-of-mouth referrals and will provide households with access to a wider pool of verified and rated workers within Tayabas City.
- **Domestic Helpers and Skilled Workers** – Domestic helpers and skilled workers will benefit from having professional digital profiles where they can present their qualifications, skills, and service rates to potential employers. The platform will provide equal access to job opportunities, help protect workers from underpricing through mandatory rate disclosure, and reward quality service through the reputation scoring system. Verified workers will also be prioritized in smart matching recommendations, encouraging workers to undergo verification and maintain high-quality service.
- **System Administrators** – System administrators will benefit from a comprehensive administrative dashboard that allows them to monitor platform activities, verify user accounts, and manage system operations. The analytics dashboard will display employment activity, geographic service distribution across barangays, service rate trends, and workforce participation data to support informed decision-making and community workforce planning.
- **Local Community** – The system will contribute to a more organized local labor market by promoting fair employment practices and improving employment accessibility within Tayabas City.

- **Researchers and Future Developers** – The findings and system design may serve as a reference for future studies and technology initiatives aimed at improving digital employment platforms in local communities.

The development of this system is motivated by the need to address the limitations of traditional hiring practices within local communities. Informal employment arrangements often lead to inefficient job matching, limited transparency, and the absence of proper verification for both workers and households. By introducing a centralized digital platform, the proposed system seeks to modernize the local employment process, improve access to job opportunities, and create a more organized and reliable employment platform for the Tayabas community.

Theoretical Framework

The development of digital employment platforms has become increasingly important in improving the way job seekers and employers connect within local communities. Traditional hiring practices in many areas still rely on informal referrals, word-of-mouth recommendations, and personal networks, which often result in limited job opportunities and inefficient matching between workers and employers. Modern information systems provide technological solutions that can address these challenges by organizing employment information, facilitating communication, and improving the overall hiring process. Web platforms, in particular, enable users to access services through internet-enabled devices, allowing job seekers to find employment opportunities and employers to identify qualified workers more efficiently. These systems may also integrate intelligent features such as automated matching, reputation tracking, and notification services to enhance user experience and system reliability. The integration of such technologies can significantly improve transparency, efficiency, and trust in employment platforms, particularly in local communities where organized labor systems are limited (Laudon & Laudon, 2022).

According to Davis (1989), the Technology Acceptance Model (TAM) explains how users come to accept and utilize a new technology. The model suggests that two major factors influence technology adoption: perceived usefulness and perceived ease of use.

Perceived usefulness refers to the degree to which a person believes that using a system will enhance job performance, while perceived ease of use refers to the degree to which a person believes that the system will be free from effort. Systems that are easy to understand and beneficial to users are more likely to be accepted and adopted. In relation to the proposed system, the Technology Acceptance Model supports the development of a user-friendly web and mobile employment platform that allows households and workers to easily navigate the system. The inclusion of simple registration processes, intuitive interfaces for both web and mobile, and email notifications helps ensure that users can interact with the platform effectively across different devices.

Blau (1964) introduced the Social Exchange Theory (SET), which explains that human interactions are based on the exchange of benefits between individuals. According to this theory, relationships are formed and maintained through trust, reciprocity, and mutual benefits. In many social systems, individuals evaluate interactions based on the rewards and costs associated with them. In digital platforms, trust is often established through feedback mechanisms and repeated interactions among users. In the context of the proposed employment system, Social Exchange Theory supports the implementation of a reputation scoring and feedback mechanism. Through this feature, households can rate workers based on their service quality, while workers can also evaluate households based on their experience. This reciprocal feedback process helps create accountability, encourages responsible behavior, and gradually builds trust among users within the platform.

The Diffusion of Innovations Theory (DOI) proposed by Rogers (1962) explains how new technologies and innovations spread within a social system over time. According to Rogers (1962), the adoption of innovations is influenced by several factors, including relative advantage, compatibility with existing practices, simplicity of use, trialability, and observability. These factors determine how quickly individuals or communities adopt new technologies. In relation to the proposed system, the Diffusion of Innovations Theory supports the design of a platform that is compatible with existing technological habits of the community. The development of a mobile application component ensures that users who primarily rely on smartphones — which are widely accessible in Tayabas City — can adopt and sustain usage of the platform more easily, thereby increasing the likelihood of successful diffusion within the local community.

The proposed system is also supported by the Job Search Theory, originally introduced by Stigler (1961) and later expanded by Diamond (1982) and Mortensen (1986) as part of the Search and Matching Theory in labor economics. This theory explains how workers and employers search for each other in labor markets where information is often incomplete or imperfect. Job seekers typically spend time and resources searching for suitable employment opportunities, while employers invest effort in finding qualified candidates. These search processes can result in delays, inefficiencies, and mismatches between job requirements and worker qualifications. In relation to this study, Job Search Theory supports the implementation of the smart matching feature of the proposed employment system. By automatically recommending workers to households based on skills, availability, verification status, and location, the system reduces the time and effort required for both parties to find suitable employment opportunities. This automated matching process helps improve efficiency in the local labor market and provides a more structured and organized employment connection within the Tayabas community.

Conceptual Framework

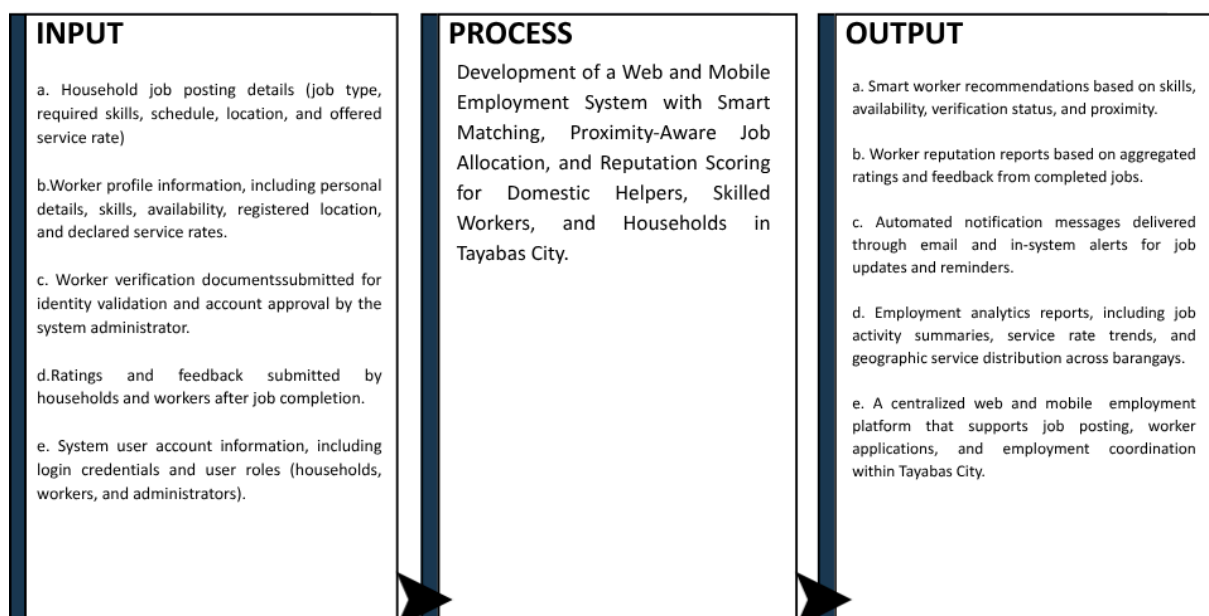


Figure 1. The Conceptual Framework of the Proposed Study

Figure 1 shows the Input–Process–Output (IPO) model, which illustrates the conceptual framework of the proposed employment platform designed to connect households with domestic helpers and skilled workers in the City of Tayabas. The framework identifies the necessary data inputs, the system development process, and the expected outputs that will result from the implementation of the proposed system. It presents the overall structure of the platform and demonstrates how the system will transform user-provided data into useful information that supports employment coordination within the community.

As shown in the model, the Input consists of the data and information required for the system to function effectively. These inputs include household job posting details, worker profiles containing skills and availability, verification documents submitted for identity confirmation, location information, ratings and feedback provided after job completion, and system user account information including login credentials and user roles. These inputs serve as the primary data used by the system in facilitating employment coordination and platform management. The Process represents the development of the proposed system entitled "Development of a Web and Mobile Employment System with Smart Matching, Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled Workers and Households in Tayabas City." During this stage, the system is designed and implemented to process the collected input data and support employment interactions between households and workers. Finally, the Output represents the results produced by the system after processing the input data. These outputs include worker recommendations based on matching criteria, reputation reports generated from user ratings and feedback, automated notifications for job updates and reminders, and employment analytics reports summarizing job activities and service distribution across barangays. Ultimately, the system will produce a centralized web and mobile employment

platform that enhances transparency, efficiency, and accessibility in connecting households with domestic helpers and skilled workers in Tayabas City.

Definition of Terms

This section presents the key terms used in the study to ensure clarity, consistency, and a common understanding among readers. The terms are classified into Technical Terms and Operational Terms based on their usage within the study.

Technical Terms

Distance Indicator – A measurement that shows the geographic distance between two locations. In this study, it refers to the system feature that displays the approximate driving distance between a worker's registered location and the household's job site using map routing services, to help households assess worker proximity and travel feasibility.

Email Notification – An electronic message sent through an email service to inform users of important events and updates. In this study, it refers to automated email alerts sent by the platform to inform users about important events such as job alerts, application updates, confirmations, and feedback reminders.

Employment Analytics Dashboard – A digital interface that visually presents employment-related data and statistics. In this study, it refers to the administrative dashboard that displays information such as job activity, verified users, geographic service distribution across barangays, and service rate summaries to support monitoring and decision-making.

Mobile Application – A software application designed to run on mobile devices such as smartphones and tablets. In this study, it refers to the mobile component of the proposed platform that allows households and workers in Tayabas City to access employment features, manage their accounts, and receive notifications through their mobile devices.

Smart Matching – An automated process that recommends or pairs users based on predefined criteria. In this study, it refers to the system feature that suggests suitable workers to households based on factors such as skills, availability, verification status, and geographic proximity to the job location.

Verification – The process of confirming the authenticity or validity of a person's identity or credentials. In this study, it refers to the administrative procedure where system administrators review and approve worker identification documents, allowing verified workers to receive a verification badge and higher priority in smart matching results.

Web Employment System – A software application that can be accessed through a web browser using an internet connection. In this study, it refers to the proposed platform that allows households, domestic helpers, and skilled workers in Tayabas City to interact, post jobs, apply for work, and manage employment activities online.

Operational Terms

Default Rating – A system-assigned rating automatically given when a household does not submit a rating within 48 to 72 hours after job completion. In such cases, the system assigns a five-star rating to prevent workers from being penalized due to missing rating and feedback.

Domestic Helper – An individual employed to perform household-related tasks and services within a residence. In this study, it refers to a registered platform user who offers services such as cleaning, cooking, laundry, childcare, and other home-based tasks to households in Tayabas City.

Household – A group of individuals living together in a residence. In this study, it refers to a registered user representing a family or individual residing in Tayabas City who can post job requests and hire domestic helpers or skilled workers through the system.

Job Posting – An announcement or listing of available work opportunities. In this study, it refers to a job listing created by a household on the platform specifying the type of service needed, required skills, schedule, location, and offered service rate.

Reputation Scoring System – A method used to evaluate credibility or performance based on ratings and feedback. In this study, it refers to the platform feature that allows households to rate workers after job completion and enables both parties to provide feedback, which is then aggregated and displayed on worker profiles to promote trust and accountability.

Skilled Worker – An individual who possesses specialized knowledge or technical expertise in a particular field. In this study, it refers to a registered user of the platform who provides technical services such as electrical work, plumbing, carpentry, or other repair services to households in Tayabas City.

Chapter2

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This portion focuses on related literature, studies, and systems in international and local settings. The researchers gathered information through journals, articles, and other forms of media available on the Internet that are relevant to the proposed study. The reviewed materials provide insights and justification for the development of the proposed Web and Mobile Employment System with Smart Matching, Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled Workers, and Households in Tayabas City.

Employment Platforms and Digital Recruitment Systems

The rapid growth of digital labor platforms has fundamentally transformed the way workers and employers connect, particularly in developing countries. The International Labour Organization (ILO, 2021) in its flagship report, *World Employment and Social Outlook 2021: The Role of Digital Labour Platforms in Transforming the World of Work*, documented how the number of digital labor platforms worldwide increased fivefold over the past decade. The report highlighted that in developing countries such as the Philippines, digital employment platforms are considered a significant source of income-generating opportunities for marginalized groups including women, youth, and persons with disabilities. Despite this global growth, the ILO emphasized that localized platforms addressing the socioeconomic conditions of municipalities and semi-urban areas remain insufficient, particularly in communities where informal hiring arrangements continue to dominate.

In the Philippine context, Soriano (2021) examined digital labor brokerage in the Philippines through a four-year digital ethnography published in *Media International Australia*. The study found that the Philippine labor market remains largely characterized by informal employment arrangements where workers and employers depend on personal referrals, social networks, and word-of-mouth to find job opportunities. Soriano observed that while digital platforms are increasingly accessible to Filipino workers, many communities — particularly those outside Metro Manila — continue to rely on informal hiring structures that lack standardization, worker verification, and employment security.

At-Your-Service, a mobile application developed for skilled workers in the Philippines and presented at an IEEE conference (Basbas et al., 2019), was designed to connect skilled workers with employers through a digital platform that improved on traditional manpower agency processes. The system allowed workers to register profiles and enabled employers to post available jobs, providing a structured alternative to informal hiring in local communities.

LocalLink, a mobile-based application for local services in barangay communities in the Philippines, was developed and evaluated with 55 participants (Supo et al., 2024). The platform aimed to connect service providers and users within localized communities, formalize the informal economy, and enhance marketplace transparency. Evaluation results showed a grand mean rating of 4.87, indicating that the platform effectively met user needs and improved community connectivity.

Smart Matching and Recommendation Systems in Employment Platforms

Möhlmann, Zalmanson, Henfridsson, and Gregory (2021) published a study in *MIS Quarterly* examining algorithmic management on online labor platforms. The researchers

found that platforms use algorithms along two dimensions — matching and control — to assign work to available workers in real time based on skills, availability, and location. The study, which drew from a qualitative examination of Uber's algorithmic systems, demonstrated that algorithmic matching reduces the need for manual coordination and enables platforms to efficiently process large volumes of job assignments. The findings emphasize that proximity-aware algorithmic matching is a practical and scalable solution for employment platforms serving local communities where manual job coordination would be inefficient.

Nikolaou (2021), in an article published in *The Spanish Journal of Psychology*, reviewed how modern recruitment practices increasingly rely on automated technologies to identify qualified candidates. The study explained that smart recruitment technologies enable organizations to screen and match candidates more effectively through automated processes, reducing the workload of recruiters while improving the accuracy of job matching results. Nikolaou emphasized that technology-driven matching systems are particularly valuable in contexts where large volumes of applications need to be efficiently processed and paired with suitable job opportunities.

The Smart Employment Platform, a system presented in the *International Journal of Multidisciplinary Research* (2025), implemented a content-based filtering algorithm that recommends jobs to workers based on skills, location, and type of work. The platform incorporated proximity-based job allocation alongside employer verification and a secure transaction model, demonstrating how intelligent matching and location awareness can be integrated into a single employment system for blue-collar and skilled workers.

Sanninganavar et al. (2021) proposed a digital skill-based job-matching system that prioritizes practical abilities over formal qualifications, featuring structured worker

registration, detailed skill profiling, employer job postings, and an automated matching engine. The system was evaluated in both rural and urban contexts and was found to improve worker visibility and reduce unemployment by efficiently matching workers to available job opportunities based on skills and proximity.

Reputation and Rating Systems in Online Service Platforms

Hesse, Teubner, and Adam (2023), in a study published in *Electronic Markets*, examined how reputation signals influence user trust and hiring decisions in gig economy platforms. Using an online experiment involving 180 actual clients of five gig economy platforms, the researchers found that workers with higher aggregated ratings are hired more frequently and are generally perceived as more reliable. The study concluded that reputation systems based on verified completed transactions are among the most powerful tools for establishing trust in digital labor markets, supporting the value of integrating mutual reputation scoring in employment platforms where both parties can rate each other.

Dellarocas (2003), in a widely cited article published in *Management Science*, examined the design and implications of online reputation mechanisms. The study explained that feedback mechanisms encourage service providers to maintain high service quality because their reputation directly affects future opportunities. Dellarocas emphasized that reputation systems transform informal word-of-mouth trust into a scalable digital mechanism that helps users make informed decisions when selecting service providers in online environments.

Resnick and Zeckhauser (2002), through an empirical analysis of eBay's reputation system published in *The Economics of the Internet and E-Commerce* (Elsevier), demonstrated that reputation profiles are predictive of future performance and that

feedback mechanisms encourage trustworthy behavior between buyers and sellers. The study found that reputation systems are effective at reducing uncertainty and promoting cooperation between parties who are unfamiliar with each other — a key requirement in community-based employment platforms where households and workers may have no prior relationship.

The LocalLink platform (Supo et al., 2024) also integrated a feedback and rating component that allowed users to evaluate service providers after each transaction. The results showed that the reputation feature improved transparency and allowed users to more easily identify reliable service providers based on accumulated community feedback.

Employment Analytics and Monitoring Dashboards in Digital Platforms

Aggarwal (2016), in the textbook *Recommender Systems: The Textbook* published by Springer, discussed how data analytics and monitoring tools within digital platforms allow administrators to examine patterns in user activity, evaluate system performance, and generate actionable insights. Aggarwal explained that data visualization within platforms transforms raw system data into meaningful summaries that support the effective management and continuous improvement of digital services — including employment platforms where monitoring job activity, user engagement, and service distribution is critical for platform governance.

The Institute for Labor Studies of the Department of Labor and Employment in the Philippines (ILS-DOLE, 2023), in its research report *Facilitating Employment Opportunities in the Digital Space*, studied the activities and services of Digital Labor Market Intermediaries (DLMIs) in the Philippines. The report found that data-driven monitoring of employment activity — including job postings, registered users, and job vacancies — was essential for

understanding employment patterns in local communities and for informing labor policy. The study highlighted that platforms which maintain structured employment data are better positioned to serve the workforce planning needs of communities.

RecruitScope, a visual analytics system developed by researchers from Huazhong University of Science and Technology (Wang et al., 2025), was designed to support multidimensional exploration of recruitment data for job seekers, employers, and HR specialists. The system used coordinated visualizations to enable users to analyze job positions and salary patterns from multiple perspectives, identify industry dynamics at the macro level, and discover high-demand roles at the micro level. Case studies demonstrated that the platform successfully revealed regional salary distribution patterns and industry growth trajectories through its analytics dashboard.

The Smart Employment Platform (2025) also incorporated an administrative monitoring component that allowed platform managers to track job postings, applications, and employer activity in real time. The monitoring dashboard provided administrators with an overview of platform usage and employment outcomes, enabling data-driven management decisions and continuous improvement of the system.

Mobile Application Development in Employment and Service Platforms

Hoehle and Venkatesh (2015), in a study published in MIS Quarterly, developed a comprehensive conceptualization of mobile application usability and demonstrated that applications designed with intuitive navigation, responsive layouts, and context-aware features are significantly more likely to be adopted and regularly used. The researchers emphasized that mobile application usability is a critical factor in determining sustained usage among diverse user groups, including those with limited prior experience with digital

services. These findings support the development of a mobile application component as a core part of the proposed employment system to ensure accessibility and adoption among households and workers in Tayabas City.

Soriano (2021), in her study of digital labor in the Philippines published in *Media International Australia*, observed that Filipino workers — particularly those in semi-urban and rural communities — increasingly access digital services through mobile devices rather than desktop computers. The study noted that mobile-accessible platforms are more inclusive and better positioned to reach workers who may not have access to traditional desktop computing environments, underscoring the importance of mobile application development for employment platforms targeting community-level users in the Philippines.

At-Your-Service (Basbas et al., 2019), developed as an Android mobile application for skilled workers in the Philippines, demonstrated how a mobile-first approach to employment platforms enables workers to register profiles, search for job opportunities, and connect with employers directly through their smartphones. The system showed that mobile applications are practical and accessible tools for delivering employment services to workers in local Philippine communities who rely on mobile devices as their primary means of digital access.

LocalLink (Supo et al., 2024), a mobile-based application for local services in barangay communities, further demonstrated that mobile employment platforms can effectively serve hyperlocal communities by connecting service providers and users within a defined geographic area. The platform's strong evaluation results confirmed that mobile-accessible employment systems are well-received by community-level users in the Philippines.

Synthesis

The reviewed literature and systems presented in this chapter include both local and international sources related to digital employment platforms, smart matching and proximity-aware job allocation, reputation and rating systems, employment analytics dashboards, and mobile application development. While these existing studies and systems demonstrate the value of digital employment platforms in improving recruitment processes and providing convenient access to employment opportunities, a critical examination of the available literature reveals significant gaps that justify the development of the proposed system.

Existing employment platforms reviewed in this chapter — including At-Your-Service (Basbas et al., 2019), LocalLink (Supo et al., 2024), and the Smart Employment Platform (2025) — address some aspects of digital employment matching but are either limited in scope, focused on specific worker categories, or designed without the combination of proximity-aware job allocation, mutual reputation scoring, worker verification, and barangay-level employment analytics that the proposed system integrates. National-scale platforms and corporate recruitment systems reviewed in the international literature are similarly not designed to address the informal hiring practices and localized employment challenges specific to communities like Tayabas City, where domestic helpers and skilled workers continue to rely on word-of-mouth referrals and personal networks.

The absence of a community-specific, multi-feature employment platform for Tayabas City — one that is accessible through both web and mobile channels, supports administrative worker verification, provides localized employment analytics, and promotes mutual accountability through reputation scoring — demonstrates that the proposed system addresses a genuine and unmet need in the local employment landscape. The proposed

system fills this gap by bringing together proximity-aware smart matching, mutual reputation scoring, worker verification, barangay-level employment analytics, and a mobile application into one platform that is specifically built for the employment needs of households, domestic helpers, and skilled workers in Tayabas City. Through the integration of these technologies, the proposed system aims to provide a reliable, organized, and inclusive platform that connects households with domestic helpers and skilled workers in Tayabas City, filling a gap that existing national and international platforms have not addressed.

Chapter3-draft

CHAPTER 3 - FORMAT

System Development Life Cycle / Project Management Framework

Requirements Analysis

Functional Requirements

Non-Functional Requirements

Data Gathering Procedure and Techniques

Research Design and Instrument

Research Design

Research Instrument

Requirement Modeling

- Context Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- Database Structure & Dictionary: Table definitions, data types, and schemas.

System Design and Architecture

Interface Diagram (Low Fidelity Wireframes)

Network Plan(optional)

Technical Background

Server-Side Requirements:

Operating System

Backend Language/Framework

Database Management System

API Integration Tools

Client-Side Requirements:

Frontend Technologies

Browser Compatibility

Hardware Minimum Specs

Development Tools:

Design/Prototyping

Version Control

AI Assistance

Statistical Treatment**ISO/IEC/IEEE 25010 Evaluation**

Agile Cycle

CHAPTER III

METHODOLOGY

This portion of the study discusses the methodology used in the development of GawaGo: A Web and Mobile App Employment Platform with Smart Matching, Reputation Scoring, and Notification System for Domestic Helpers, Skilled Workers, and Households in Tayabas City. It covers the software development methodology, the phases of development followed by the research team, and the procedures applied in designing, building, testing, and deploying the proposed system.

Software Development Lifecycle

AGILE ITERATION PROCESS DIAGRAM

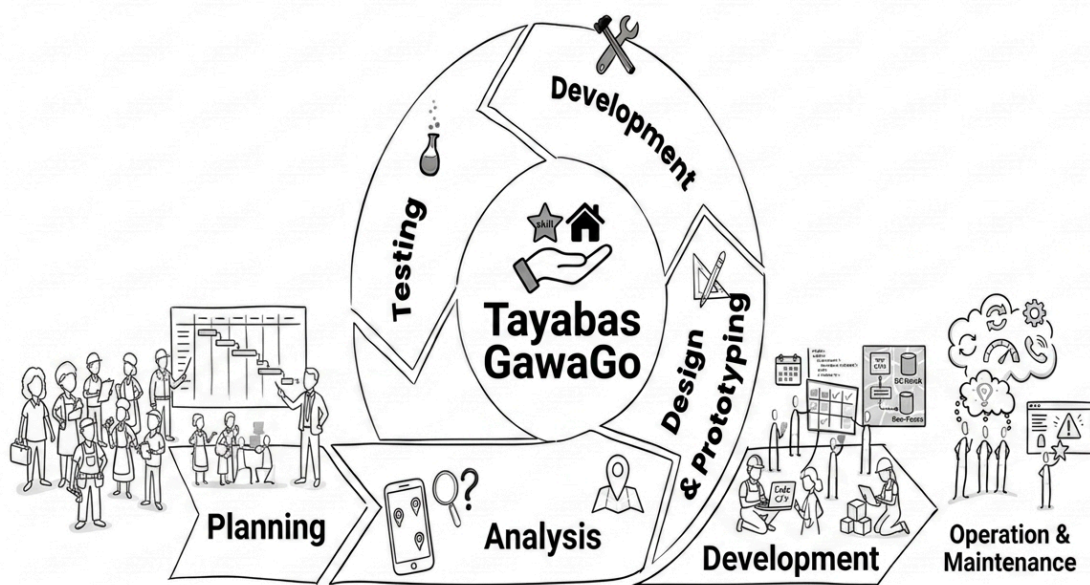


Figure 1. Agile Software Development Methodology of GawaGo

Figure 1 illustrates the Agile Software Development Lifecycle adopted in the development of the GawaGo Employment Platform. It shows the sequential phases of the system development process, which include Planning, Requirements Analysis, System Design, Development, Testing, Deployment, and Maintenance and Improvement. These

phases guide the researchers in systematically designing, building, evaluating, and maintaining the proposed system.

Planning: The Planning Phase focuses on setting the project objectives, scope, target users, and system features. The researchers identified employment-related problems in Tayabas City and defined system modules such as user management, job posting, applications, smart matching, verification, notifications, ratings, and analytics. A development schedule was also prepared.

Requirements Analysis: This phase involves gathering system requirements through interviews, consultations, observations, and related studies. The researchers identified the needed inputs, processes, outputs, and system requirements. Security, usability, and performance requirements were also defined based on ISO/IEC 25010 standards.

System Design: In this phase, the system structure was created, including database design, user interface layout, diagrams, and wireframes for both web and mobile versions. The smart matching system, notifications, rating system, and analytics dashboard were also designed with a focus on usability and accessibility.

Development: This phase involves the actual coding of the system using Agile iterations. Features such as registration, login, profiles, job posting, applications, smart matching, verification, ratings, notifications, and analytics were developed step by step. The web and mobile applications were connected to a shared Django backend API to ensure consistent data across platforms.

Testing: The system was tested in terms of functionality, performance, security, usability, and reliability. Testing methods included unit testing, integration testing, system testing, and user

acceptance testing (UAT). Errors found during testing were fixed, and feedback from users was used to improve the system.

Deployment: After testing, the system was deployed for actual use. Accounts for households, workers, and administrators were created, and the system was monitored during initial use to ensure smooth operation. Any issues found during this stage were recorded for correction.

Maintenance and Improvement: This phase involves ongoing monitoring and improvement of the system. Errors are fixed, features are improved, and performance is optimized based on user feedback. Future enhancements such as payment features, additional notifications, and advanced analytics may also be added to improve the system further.

RequirementsAnalysis

Requirements Analysis

Functional Requirements

The functional requirements of the proposed Web and Mobile Employment System with Smart Matching, Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled Workers and Households in Tayabas City include the specific features and functionalities that the system must perform to support its intended operations.

- User Registration and Authentication - The system shall allow households, workers, and administrators to register and securely log in using validated credentials.
- Household Profile Management - The system shall enable households to create, view, and update profile information, including contact details, location, and hiring preferences.
- Worker Profile Management - The system shall allow workers to manage professional profiles containing skills, experience, availability, service rates, and verification documents.
- Job Posting Management - The system shall allow households to create and manage job postings containing job details, required skills, schedules, locations, and service rates.
- Job Application Management - The system shall allow workers to browse job listings, submit applications, and track application status.
- Applicant Review and Worker Selection - The system shall enable households to review applications, evaluate worker profiles, and confirm workers for job assignments.

- Smart Worker Matching - The system shall recommend and rank workers based on skill compatibility, availability, verification status, and approximate geographic distance while allowing households to make the final hiring decision.
- Distance Indicator - The system shall display the estimated straight-line distance between the worker's registered location and the job location.
- Service Rate Visibility - The system shall display worker service rates and maintain visibility of agreed rates within job records.
- Worker Verification Management - The system shall allow workers to submit verification documents and enable administrators to approve or reject verification requests.
- Ratings and Reviews - The system shall allow households and workers to provide ratings and reviews after job completion.
- Automatic Default Rating - The system shall automatically assign a default five-star rating to workers when household feedback is not submitted within 48–72 hours after job completion.
- New User Rating Display -The system shall display “No Ratings Yet” for users without existing reviews.
- Notifications - The system shall provide email and in-system notifications for application updates, worker confirmations, verification results, feedback reminders, and other important activities.
- Employment Analytics Dashboard - The system shall provide administrators with summarized employment statistics, including job postings, applications, completed services, verification records, and workforce trends.

- Administrative Management and Monitoring - The system shall allow administrators to manage users, monitor platform activities, process verifications, access analytics, and maintain system operations.

Non-Functional Requirements

The non-functional requirements of the proposed Web and Mobile Employment System with Smart Matching, Proximity-Aware Job Allocation, and Reputation Scoring for Domestic Helpers, Skilled Workers and Households in Tayabas City include the quality attributes that define how the system performs its functions.

- Usability - The system shall provide a simple, intuitive, and user-friendly interface that enables users to perform tasks efficiently.
- Responsiveness - The system shall adapt seamlessly across desktop computers, tablets, and smartphones.
- Security - The system shall implement secure authentication, password protection, and role-based access control to safeguard user accounts and data.
- Data Privacy - The system shall protect personal and verification information and ensure that data is used only for authorized platform purposes.
- Reliability - The system shall operate consistently and accurately with minimal errors or service interruptions.
- Performance - The system shall process requests and display information within acceptable response times, including smart matching and analytics functions.
- Scalability - The system shall support future growth in users, records, service categories, and platform features.
- Maintainability - The system shall be structured to allow efficient updates, modifications, and troubleshooting.

- Compatibility - The system shall provide consistent functionality across supported web browsers and mobile devices.
- Availability - The system shall remain accessible to users whenever internet connectivity is available, with minimal downtime.
- Accuracy of Matching and Distance Display - The system shall provide accurate worker recommendations and display approximate distance information based on registered locations.

DataGathering and Techniques

Data Gathering Procedure and Techniques

The researchers used several data gathering procedures and techniques to collect the information needed for the development of the proposed system entitled “GawaGo: A Web and Mobile Employment Platform with Smart Matching, Reputation Scoring, and Notification System for Domestic Helpers, Skilled Workers, and Households in Tayabas City.” The collected data served as the basis for identifying system requirements, user needs, existing challenges, and design considerations for the proposed platform.

The following data gathering techniques were used in the study:

Client Consultation and Interviews

The researchers conducted consultation meetings and interviews with the project stakeholders, particularly representatives from the City Planning Office and the Head of Information Technology of Tayabas City. These consultations were conducted to gather information about the goals, scope, requirements, and expected features of the proposed system. The discussions focused on employment-related challenges, worker verification procedures, communication methods, service accessibility, and other important features needed for the platform. The information gathered helped the researchers identify the needs and expectations of the stakeholders and served as a basis for the system requirements.

Observation of Existing Employment Practices

The researchers observed the existing employment and hiring practices commonly used within the community. These included referrals, social media postings, word-of-mouth hiring, and direct communication between households and workers. Through observation, the researchers gained a better understanding of current processes, identified existing

problems and limitations, and recognized opportunities for improving employment coordination, transparency, and accessibility through the proposed system.

Review of Related Literature and Existing Systems

The researchers reviewed related studies, journals, articles, and existing employment platforms related to digital employment systems, worker verification, reputation management, notification systems, employment analytics, and smart matching technologies. References related to software development, web technologies, mobile applications, and database systems were also examined. This review helped the researchers understand existing technologies, identify useful features and best practices, and support the planning and design of the proposed platform.

Analysis of Gathered Data and Requirement Identification

The researchers analyzed the information gathered from interviews, observations, and literature reviews to identify the needs, challenges, and expectations related to employment coordination in Tayabas City. The analysis focused on employment accessibility, worker verification, communication efficiency, service rate transparency, worker recommendations, reputation management, notifications, and employment monitoring. The results of the analysis served as the basis for defining the system's functional and non-functional requirements, database structure, process flow, and user interface design. The gathered information also guided the development of a reliable, organized, and user-friendly employment platform that addresses the needs of households, workers, and administrators.

Research Design and Instrument

Research Design and Instrument

Research Design

This study will use a Descriptive-Developmental Research Design in the development of the proposed GawaGo Employment Platform. The descriptive aspect of the study focuses on identifying and analyzing employment-related practices and challenges within Tayabas City through stakeholder consultations, observations, and document reviews. This includes issues related to job accessibility, worker availability, verification concerns, communication difficulties, service rate transparency, and the use of informal hiring methods such as referrals and word-of-mouth recommendations.

The information gathered will help the researchers understand the needs and expectations of households, workers, and administrators, and identify the system features required to address existing problems. The developmental aspect focuses on the design, development, implementation, and evaluation of the proposed system entitled “GawaGo: A Web and Mobile Employment Platform with Smart Matching, Reputation Scoring, and Notification System for Domestic Helpers, Skilled Workers, and Households in Tayabas City.”

This process includes designing the system architecture, database, and user interfaces; developing the web and mobile platform; implementing smart matching, reputation scoring, verification management, and notification features; and evaluating the system using the ISO/IEC 25010 Software Quality Standards. The Descriptive-Developmental Research Design is appropriate for this study because it allows the researchers to understand existing employment-related challenges and develop a technology-based solution that addresses the identified needs of the community.

Research Instrument

The researchers will use several research instruments to gather the data needed for the development and evaluation of the proposed GawaGo Employment Platform.

Interview Guides

The researchers will prepare interview guides with structured questions for meetings and interviews with stakeholders, especially representatives from the City Planning Office and the Head of Information Technology of Tayabas City. The questions will focus on employment problems, worker verification needs, communication needs, system features, administrative needs, and suggestions to improve employment coordination. The answers will be used to identify system requirements and user needs.

Observation Checklists

Observation checklists will be used to record current hiring and job-seeking practices in the community. These will help identify common work processes, referral methods, communication methods, user behavior, and problems during hiring. The observations will support the analysis of current employment practices and guide the design of the proposed system.

System Evaluation Forms

System evaluation forms will be used to check the quality and effectiveness of the system after development and testing. These will be given to selected users and evaluators and will be based on the ISO/IEC 25010 Software Quality Standards. The evaluation will include areas such as functional suitability, usability, reliability, performance, security, compatibility, maintainability, user satisfaction, and usefulness. The results of the evaluation will help determine whether the system meets the needs and expectations of its intended

users. Overall, these research instruments will provide the information needed for the design, development, improvement, and evaluation of the proposed GawaGo Employment Platform.

Context Diagram

DFD.

Figure 2. The Data Flow Diagram of the Proposed Employment System

ERD.

DataSchema

GawaGo Database Structure and Dictionary

This section presents the database structure and dictionary of the GawaGo Employment Platform. The format follows a table dictionary style showing the column number, field name, data type, nullability, default value, and primary key indicator. The database structure is based on the current Django backend implementation of the system.

Table 1. auth_user

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	1	NULL	1
1	password	VARCHAR(128)	1	NULL	0
2	last_login	DATETIME	0	NULL	0
3	is_superuser	BOOLEAN	1	NULL	0
4	username	VARCHAR(150)	1	NULL	0
5	first_name	VARCHAR(150)	1	NULL	0
6	last_name	VARCHAR(150)	1	NULL	0
7	email	VARCHAR(254)	1	NULL	0
8	is_staff	BOOLEAN	1	NULL	0
9	is_active	BOOLEAN	1	NULL	0
10	date_joined	DATETIME	1	NULL	0

This table stores the login and account information of registered users in the GawaGo Employment Platform. It includes user credentials, name, email address, account status, and administrator access indicators. Household and worker-specific profile information is stored separately in the accounts_userprofile table.

Table 2

accounts_userprofile

cid	name	type	notnull	dflt_value	pk
-----	------	------	---------	------------	----

0	id	BIGINT	1	NULL	1
---	----	--------	---	------	---

1	user_id	INTEGER	1	NULL	0
2	role	VARCHAR(20)	1	worker	0
3	phone	VARCHAR(30)	1	"	0
4	bio	TEXT	1	"	0
5	years_experience	INTEGER UNSIGNED	1	0	0
6	skills	JSON	1	[]	0
7	hourly_rate	DECIMAL(10,2)	0	NULL	0
8	daily_rate	DECIMAL(10,2)	0	NULL	0
9	verification_status	VARCHAR(20)	1	pending	0
10	location_label	VARCHAR(255)	1	"	0
11	latitude	DECIMAL(10,7)	0	NULL	0
12	longitude	DECIMAL(10,7)	0	NULL	0
13	profile_photo	VARCHAR(100)	0	NULL	0
14	average_rating	DECIMAL(3,2)	0	NULL	0
15	rating_count	INTEGER UNSIGNED	1	0	0

This table stores the profile information of each registered user. It identifies whether the user is a household or worker and stores personal details, worker skills, service rates, verification status, location coordinates, profile image, and rating summary. Each profile is connected to one record in the auth_user table.

Table 3

accounts_workeravailability

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	worker_id	INTEGER	1	NULL	0
2	date	DATE	1	NULL	0
3	start_time	TIME	1	NULL	0
4	end_time	TIME	1	NULL	0
5	is_available	BOOLEAN	1	1	0

This table stores the availability schedule of workers. It records the date and time windows when a worker is available for job matching. The system uses this table during smart matching to recommend workers who are available for the requested schedule.

Table 4

accounts_passwordresetrequest

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	user_id	INTEGER	1	NULL	0
2	email	VARCHAR(254)	1	NULL	0
3	token_hash	VARCHAR(128)	1	NULL	0
4	created_at	DATETIME	1	NULL	0
5	expires_at	DATETIME	1	NULL	0
6	used_at	DATETIME	0	NULL	0
7	attempts	INTEGER UNSIGNED	1	0	0

This table stores password reset requests made by users. It contains the email address, hashed reset token, expiration date, usage status, and number of verification attempts. The system uses this table to securely process forgot-password and reset-password transactions.

Table 5

accounts_signupverificationrequest

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	user_id	INTEGER	1	NULL	0
2	email	VARCHAR(254)	1	NULL	0
3	token_hash	VARCHAR(128)	1	NULL	0
4	created_at	DATETIME	1	NULL	0
5	expires_at	DATETIME	1	NULL	0
6	verified_at	DATETIME	0	NULL	0

This table stores account verification requests created during user registration. It keeps the email address, hashed verification token, expiration date, and verification timestamp. This supports email verification before or after account activation depending on the registration flow.

Table 6

jobs_jobposting

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	household_id	INTEGER	1	NULL	0
2	title	VARCHAR(255)	1	NULL	0
3	job_type	VARCHAR(120)	1	NULL	0
4	required_skill	VARCHAR(120)	1	NULL	0
5	schedule	VARCHAR(255)	1	NULL	0
6	schedule_type	VARCHAR(80)	1	"	0
7	preferred_date	DATE	0	NULL	0
8	preferred_time	TIME	0	NULL	0
9	description	TEXT	1	"	0
10	location_label	VARCHAR(255)	1	NULL	0
11	latitude	DECIMAL(10,7)	1	NULL	0
12	longitude	DECIMAL(10,7)	1	NULL	0
13	service_rate	DECIMAL(10,2)	1	NULL	0
14	worker_slots	INTEGER UNSIGNED	1	1	0
15	status	VARCHAR(20)	1	open	0
16	created_at	DATETIME	1	NULL	0
17	completed_at	DATETIME	0	NULL	0

This table stores job postings created by household users. It includes the job title, type, required skill, schedule, location, service rate, number of workers needed, and job status. The system uses this table for job listing, matching, application management, and service completion tracking.

Table 7

jobs_jobapplication

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	job_id	BIGINT	1	NULL	0
2	worker_id	INTEGER	1	NULL	0
3	status	VARCHAR(20)	1	pending	0

4	applied_at	DATETIME	1	NULL	0
5	updated_at	DATETIME	1	NULL	0
6	note	TEXT	1	"	0

This table stores job applications submitted by workers. It links a worker to a specific job posting and records the application status, submission date, update date, and worker note. The system uses this table to support hiring decisions, rejection, completion, and application tracking.

Table 8

jobs_jobimage

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	job_id	BIGINT	1	NULL	0
2	image	VARCHAR(100)	1	NULL	0
3	uploaded_at	DATETIME	1	NULL	0
4	order	INTEGER UNSIGNED	1	0	0

This table stores uploaded images related to job postings. It allows households to attach images that help workers understand the job location, task condition, or service requirement. The display order is stored to control the sequence of job images.

Table 9

reviews_review

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	author_id	INTEGER	1	NULL	0
2	target_id	INTEGER	1	NULL	0
3	author_role	VARCHAR(20)	1	NULL	0
4	target_role	VARCHAR(20)	1	NULL	0
5	job_id	BIGINT	0	NULL	0
6	job_title	VARCHAR(255)	1	"	0
7	rating	DECIMAL(2,1)	0	NULL	0
8	feedback	TEXT	1	"	0

9	created_at	DATETIME	1	NULL	0
---	------------	----------	---	------	---

This table stores reviews and feedback submitted after job completion. It supports household-to-worker ratings and worker-to-household feedback. The table records the author, target user, related job, rating value, written feedback, and creation date to support the platform reputation system.

Table 10

notifications_notification

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	recipient_id	INTEGER	1	NULL	0
2	notification_type	VARCHAR(30)	1	NULL	0
3	title	VARCHAR(255)	1	NULL	0
4	message	TEXT	1	NULL	0
5	is_read	BOOLEAN	1	0	0
6	created_at	DATETIME	1	NULL	0

This table stores in-system notifications received by users. It records the notification recipient, notification category, title, message, read status, and date created. The system uses this table for job updates, application status changes, verification results, account activity, completion reminders, and review reminders.

Table 11

common_verificationrequest

cid	name	type	notnull	dflt_value	pk
0	id	BIGINT	1	NULL	1
1	worker_id	INTEGER	1	NULL	0
2	primary_id_name	VARCHAR(255)	1	NULL	0
3	secondary_doc_name	VARCHAR(255)	1	NULL	0
4	primary_id_file	VARCHAR(100)	0	NULL	0
5	secondary_doc_file	VARCHAR(100)	0	NULL	0
6	notes	TEXT	1	"	0
7	status	VARCHAR(20)	1	pending	0
8	submitted_at	DATETIME	1	NULL	0

9	reviewed_at	DATETIME	0	NULL	0
10	review_note	TEXT	1	"	0

This table stores worker verification requests submitted to the system administrator. It contains the names and uploaded files of the worker's identification documents, submission notes, verification status, submission date, review date, and administrator review remarks. This table supports the trust and verification feature of the platform.

Relationship Summary

Relationship	Cardinality	Description
auth_user to accounts_userprofile	1:1	Each user account has one profile record
auth_user to accounts_workeravailability	1:N	One worker can create many availability records
auth_user to jobs_jobposting	1:N	One household can create many job postings
auth_user to jobs_jobapplication	1:N	One worker can submit many job applications
jobs_jobposting to jobs_jobapplication	1:N	One job posting can receive many applications
jobs_jobposting to jobs_jobimage	1:N	One job posting can have many uploaded images
auth_user to reviews_review	1:N	One user can write and receive many reviews
jobs_jobposting to reviews_review	1:N	One completed job can have reviews from both parties
auth_user to notifications_notification	1:N	One user can receive many notifications
auth_user to common_verificationrequest	1:N	One worker can submit many verification requests

The database uses foreign key relationships to connect user accounts, profiles, jobs, applications, reviews, notifications, and verification records. The matching and analytics modules are implemented through backend service logic and currently compute results from the existing tables instead of storing them in separate database tables.

DATASHEMA.

auth_user	
id	int
password	varchar(128) NN
last_login	datetime
is_superuser	boolean NN
username	varchar(150) NN
first_name	varchar(150) NN
last_name	varchar(150) NN
email	varchar(254) NN
is_staff	boolean NN
is_active	boolean NN
date_joined	datetime NN

accounts_userprofile	
id	bigint
user_id	int NN
role	varchar(20) NN
phone	varchar(30) NN
bio	text NN
years_experience	int NN
skills	json NN
hourly_rate	decimal(10,2)
daily_rate	decimal(10,2)
verification_status	varchar(20) NN
location_label	varchar(255) NN
latitude	decimal(10,7)
longitude	decimal(10,7)
profile_photo	varchar(100)
average_rating	decimal(3,2)
rating_count	int NN

jobs_jobapplication	
id	bigint
job_id	bigint NN
worker_id	int NN
status	varchar(20) NN
applied_at	datetime NN
updated_at	datetime NN
note	text NN

jobs_jobimage	
id	bigint
job_id	bigint NN
image	varchar(100) NN
uploaded_at	datetime NN
order	int NN

reviews_review	
id	bigint
author_id	int NN
target_id	int NN
author_role	varchar(20) NN
target_role	varchar(20) NN
job_id	bigint
job_title	varchar(255) NN
rating	decimal(3,1)
feedback	text NN
created_at	datetime NN

accounts_workeravailability	
id	bigint
worker_id	int NN
date	date NN
start_time	time NN
end_time	time NN
is_available	boolean NN

accounts_passwordresetrequest	
id	bigint
user_id	int NN
email	varchar(254) NN
token_hash	varchar(128) NN
created_at	datetime NN
expires_at	datetime NN
used_at	datetime
attempts	int NN

accounts_signupverificationrequest	
id	bigint
user_id	int NN
email	varchar(254) NN
token_hash	varchar(128) NN
created_at	datetime NN
expires_at	datetime NN
verified_at	datetime

jobs_jobposting	
id	bigint
household_id	int NN
title	varchar(255) NN
job_type	varchar(120) NN
required_skill	varchar(120) NN
schedule	varchar(255) NN
schedule_type	varchar(80) NN
preferred_date	date
preferred_time	time
description	text NN
location_label	varchar(255) NN
latitude	decimal(10,7) NN
longitude	decimal(10,7) NN
service_rate	decimal(10,2) NN
worker_slots	int NN
status	varchar(20) NN
created_at	datetime NN
completed_at	datetime

notifications_notification	
id	bigint
recipient_id	int NN
notification_type	varchar(30) NN
title	varchar(255) NN
message	text NN
is_read	boolean NN
created_at	datetime NN

common_verificationrequest	
id	bigint
worker_id	int NN
primary_id_name	varchar(255) NN
secondary_doc_name	varchar(255) NN
primary_id_file	varchar(100)
secondary_doc_file	varchar(100)
notes	text NN
status	varchar(20) NN
submitted_at	datetime NN
reviewed_at	datetime
review_note	text NN

The Data Dictionary from Database Schema

Table 1. users

FieldName	DataType	PK	FK	Null	Description
user_id	INTEGER	1	0	0	Unique user Identifier
full_name	VARCHAR(100)	0	0	0	Full name of the user

last_name	VARCHAR(100)	0	0	0	Last name of the user
email	VARCHAR(100)	0	0	0	Email address (unique)
password	VARCHAR(255)	0	0	0	Hashed password
contact_number	VARCHAR(20)	0	0	0	Contact number
role	ENUM("Household","Worker","Administrator")	0	0	0	User role
address	VARCHAR(255)	0	0	1	Complete address
barangay	VARCHAR(100)	0	0	1	Barangay
latitude	DECIMAL(10,8)	0	0	1	Latitude coordinate
longitude	DECIMAL(11,8)	0	0	1	Longitude coordinate
created_at	DATETIME	0	0	0	Account creation datetime

The users table is the central table of the database, storing the authentication credentials, contact information, and role assignments for all registered system users including households, workers, and administrators.

Table 2. household_profiles

FieldName	DataType	PK	FK	Null	Description
household_id	INTEGER	1	0	0	Unique household profile ID
user_id	INTEGER	0	0	0	Reference to user
household_name	VARCHAR(150)	0	0	0	Name of household
address	VARCHAR(255)	0	0	0	Household address

preferred_contact	VARCHAR(50)	0	0	0	Preferred contact method(phone'email)
created_at	DATETIME	0	0	0	Profile creation datetime

The household_profiles table stores additional profile information specific to users registered under the Household role, including the household name, address, and preferred contact method.

Table 3. worker_profiles

FieldName	DataType	PK	FK	Null	Description
worker_id	INTEGER	1	0	0	Unique worker profile ID
user_id	INTEGER	0	1	0	Reference to users
bio	TEXT	0	0	1	Short biography
skills	TEXT	0	0	1	Skills (comma separated or text)
hourly_rate	DECIMAL(10,2)	0	0	1	Hourly rate
daily_rate	DECIMAL(10,2)	0	0	1	Daily rate
availability_status	ENUM("Available", "Busy", "Unavailable")	0	0	0	Current availability
verification_status	ENUM("Pending", "Verified", "Rejected")	0	0	0	Verification status
average_rating	DECIMAL(3,2)	0	0	0	Average rating (1.00 - 5.00)
created_at	DATETIME	0	0	0	Profile creation datetime
updated_at	DATETIME			1	Profile update datetime

The worker_profiles table stores professional information about users registered as workers, including their skills, service rates, availability, verification status, and aggregated reputation score.

Table 4. job_postings

FieldName	DataType	PF	FK	Null	Description
job_id	INTEGER	1	0	0	Unique job posting ID
household_id	INTEGER	0	1	0	Reference to household_profiles
job_type	VARCHAR(100)	0	0	0	Type of job
description	TEXT	0	0	0	Job description
required_skills	TEXT	0	0	1	Required skills
schedule_date	DATE	0	0	0	Date of schedule
schedule_time	TIME	0	0	0	Time of schedule
location	VARCHAR(150)	0	0	0	Job location
barangay	VARCHAR(100)	0	0	0	Latitude coordinate
latitude	DECIMAL(10,8)	0	0	1	Latitude coordinate
longtitude	DECIMAL(11,8)	0	0	1	Longitude coordinate
offered_rate	DECIMAL(10,2)	0	0	0	Offered service rate
status	ENUM("In Progress", "Completed", "Cancelled")	0	0	0	Job status
created_at	DATETIME	0	0	0	Posting creation datetime
updated_at	DATETIME	0	0	1	Last update datetime

The job_postings table records all job listings created by household users. Each record captures the type of service needed, the required skills, the schedule, the job location with geographic coordinates, the offered service rate, and the current status of the posting.

Table 5. job_applications

FieldName	DataType	PK	FK	Null	Description
application_id	INTEGER	1	0	0	Unique application ID
job_id	INTEGER	0	1	0	Reference to job_postings
worker_id	INTEGER	0	1	0	Reference to worker_profiles
application_status	ENUM('Pending', 'Reviewed', 'Accepted', 'Rejected')	0	0	0	Application status
proposed_rate	DECCIMAL(10,2)	0	0	1	Rate proposed by worker
message	TEXT	0	0	1	Message to household
applied_at	DATETIME	0	0	0	Application datetime
updated_at	DATETIME	0	0	1	Last update datetime

The job_applications table records all applications submitted by workers in response to published job postings. It tracks the application status, the rate proposed by the worker, any accompanying message, and relevant timestamps.

Table 6. job_confirmations

FieldName	Datatype	PK	FK	Null	Description
confirmation_id	INTEGER	1	0	0	Unique confirmation ID
job_id	INTEGER	0	1	0	Reference to job_postings
household_id	INTEGER	0	1	0	Reference to household_profiles

worker_id	INTEGER	0	1	0	Reference to worker_profiles
agreed_id	DECIMAL(10.2)	0	0	0	Agreed service rate
confirmation_status	ENUM('Pending', 'Confirmed', 'Cancelled', 'Completed')	0	0	0	Confirmation status
confirmed_id	DATETIME	0	0	1	Date and time confirmed
updated_at	DATETIME	0	0	1	Last update datetime

The job_confirmations table records the formal hiring agreement between a household and a selected worker for a specific job posting, including the agreed service rate and the current status of the engagement.

Table 7. reviews

FieldName	DateType	PK	FK	Null	Description
review_id	INTEGER	1	0	0	Unique review ID
job_id	INTEGER	0	1	0	Reference to job_postings
reviewer_id	INTEGER	0	1	0	Reference to users (who gives the review)
reviewee_id	INTEGER	0	1	0	Reference to users (who receives review)
rating	TINYINT	0	0	0	Rating (1 to 5)
comment	TEXT	0	0	1	Review comment
is_default_rating	BOOLEAN	0	0	0	1 = Default (auto), 0 = Manual
created_at	DATETIME	0	0	0	Review datetime

The reviews table stores ratings and written feedback submitted by households and workers following the completion of a job engagement. It also records whether a rating was submitted manually by the user or assigned automatically by the system.

Table 8. verification_records

FieldName	DataType	PK	FK	Null	Description
verification_id	INTEGER	1	0	0	Unique verification ID
worker_id	INTEGER	0	1	0	Reference to worker_profiles
document_id	VARCHAR(100)	0	0	0	Type of document
document_file	VARCHAR(255)	0	0	0	File path or link
verification_status	ENUM('Pending', 'Verified', 'Rejected')	0	0	0	Verification status
reviewed_by	INTEGER	0	1	1	Reference to users (admin who reviewed)
reviewed_at	DATETIME	0	0	1	Review datetime
created_at	DATEIME	0	0	0	Record creation datetime

The verification_records table tracks all document submissions made by workers for identity verification purposes, including the type of document submitted, the file reference, and the review decision made by the administrator.

Table 9. notifications

FieldName	DataType	PK	FK	Null	Description
notification_id	INTEGER	1	0	0	Unique notification ID
user_id	INTEGER	0	1	0	Reference to users (recipient)
title	VARCHAR(150)	0	0	0	Notification title
message	TEXT	0	0	0	Notification message
notification_type	VARCHAR(50)	0	0	0	Type (Job Update, Application, Reminder, System, etc.)
is_read	BOOLEAN	0	0	0	1 = Read, 0 = Unread

created_at	DATETIME	0	0	0	Record creation datetime
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The notifications table logs all in-app notifications generated and delivered to users across all platform workflows, including job updates, application status changes, verification results, and system-generated reminders.

Table 10. analytics_records

FieldName	DataType	PK	FK	Null	Description
analytics_id	INTEGER	1	0	0	Unique analytics ID
total_job_posting	INTEGER	0	0	0	Total job postings
active_applications	INTEGER	0	0	0	Active applications
completed_services	INTEGER	0	0	0	Completed services
cancelled_matches	INTEGER	0	0	0	Cancelled request
ongoing_matches	INTEGER	0	0	0	Ongoing matches
verified_users	INTEGER	0	0	0	Verified users
pending_verificatons	INTEGER	0	0	0	Pending verifications
created_at	DATETIME	0	0	0	Record datetime

The analytics_records table stores aggregated platform-wide employment activity metrics used by the administrative analytics dashboard for monitoring, reporting, and informed decision-making.

LOWFIDELITY.

System Design and Architecture

Interface Diagram (Low Fidelity Wireframes)

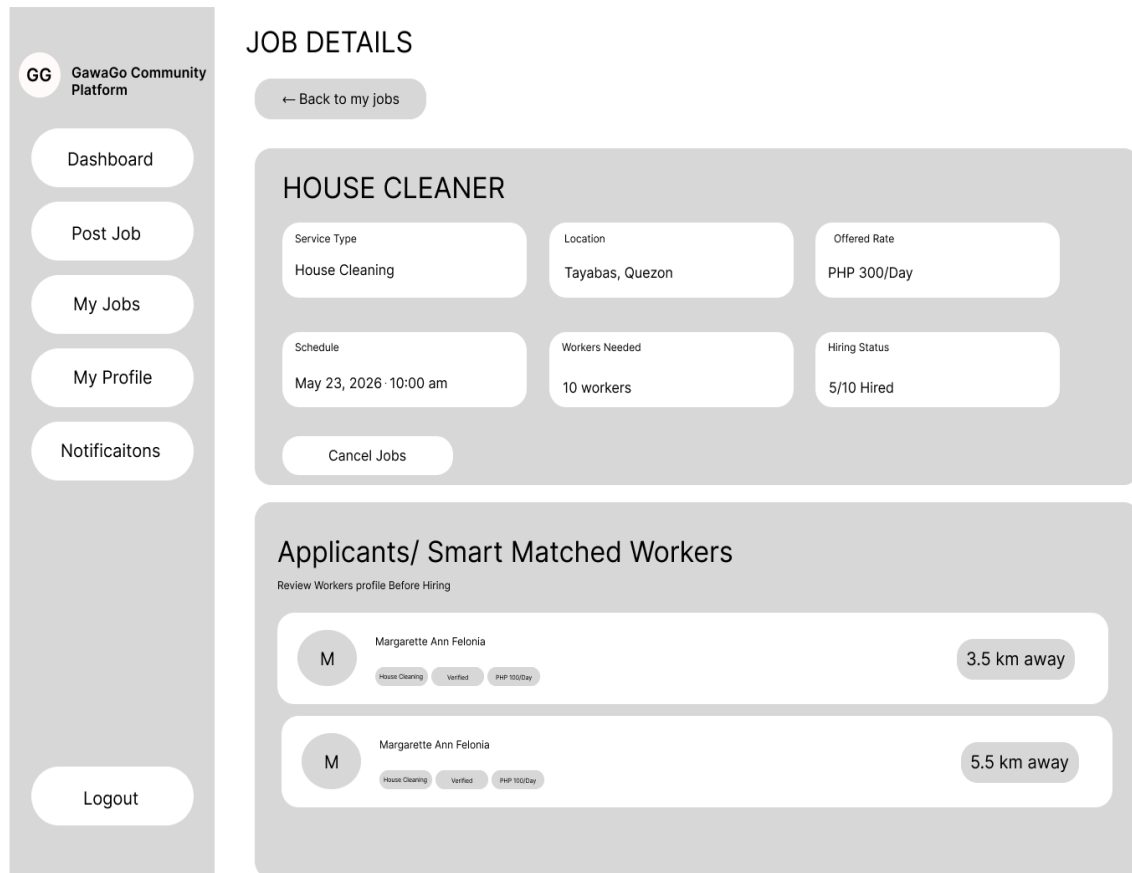


Figure X. Low-Fidelity Prototype of the Job Details and Worker Matching Interface

The figure presents the low-fidelity prototype of the Job Details and Worker Matching interface of the GawaGo Community Platform. The layout consists of a navigation panel for accessing key system features and a main content area displaying job information, including service type, location, offered rate, schedule, required workers, and hiring status. The interface also includes a worker matching section that lists applicants with their skills, verification status, ratings, and proximity to the job location. This design allows household users to review and manage job postings while evaluating potential workers before hiring.

GG

GawaGo Community Platform

Dashboard

Post Job

My Jobs

My Profile

Notificaitons

Logout

Post a New Job

Job title

Services Type

Schedule Type

Preferred Date

Preferred Time

Job Description

Barangay

Street / House Number

Workers Needed

Offered Rate

Rate Type

Reference Images

Drop Images here

Use my Current Location

Pin on map manually

Post Job

Cancel

Figure X. Low-Fidelity Prototype of the Job Posting Interface

The figure presents the low-fidelity prototype of the Job Posting interface of the GawaGo Community Platform. The design includes a navigation sidebar and a job posting form that allows household users to enter job details such as title, service type, schedule, description, location, number of workers needed, and offered rate. A map feature is integrated to enable users to specify the job location using their current location or manual pin placement. The interface also provides an image upload section for reference photos and action buttons for submitting or canceling the job posting. This design supports efficient creation and management of job requests within the platform.

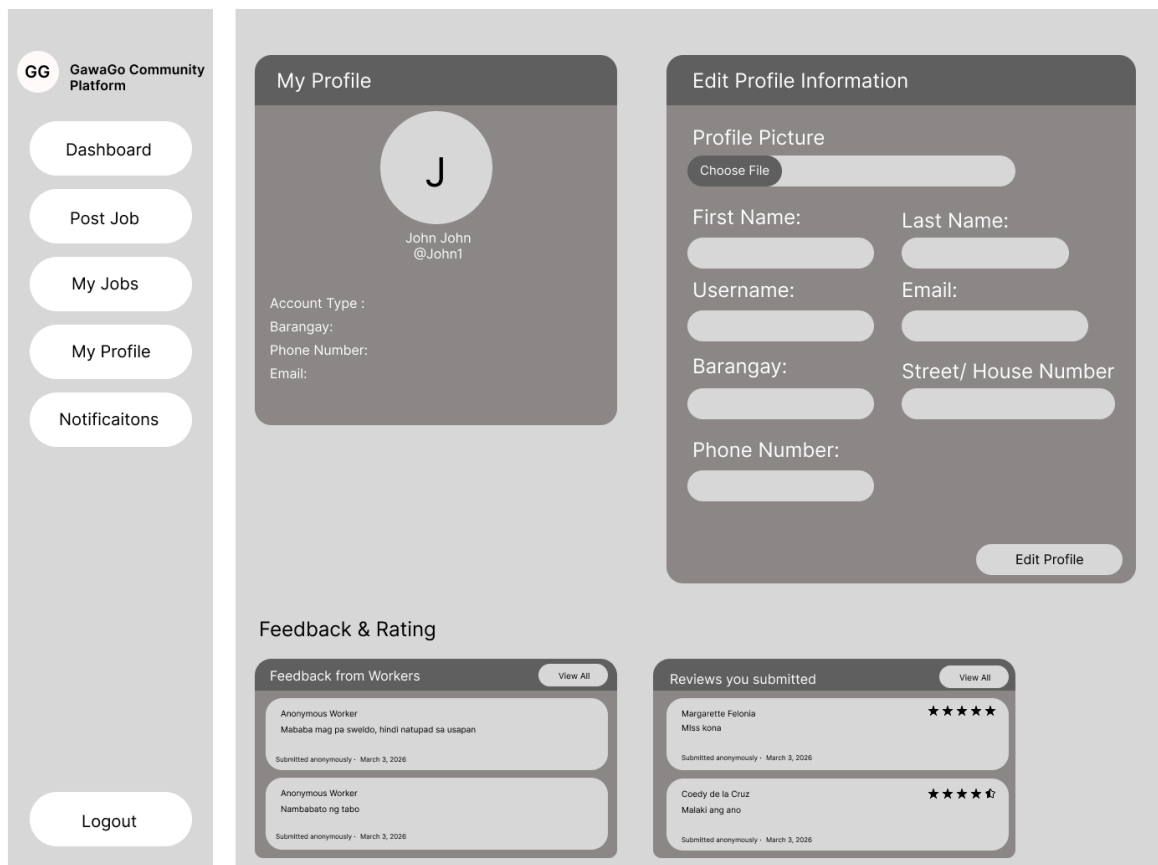


Figure X. Low-Fidelity Prototype of the User Profile and Feedback Interface

The figure presents the low-fidelity prototype of the User Profile and Feedback interface of the GawaGo Community Platform. The design includes a navigation sidebar, a profile information section, and a profile editing module that allows users to update personal details and profile pictures. The interface also provides sections for viewing feedback received from workers and reviews submitted by the user, including ratings and comments. This design supports profile management, user interaction, and transparency through a centralized feedback and review system.

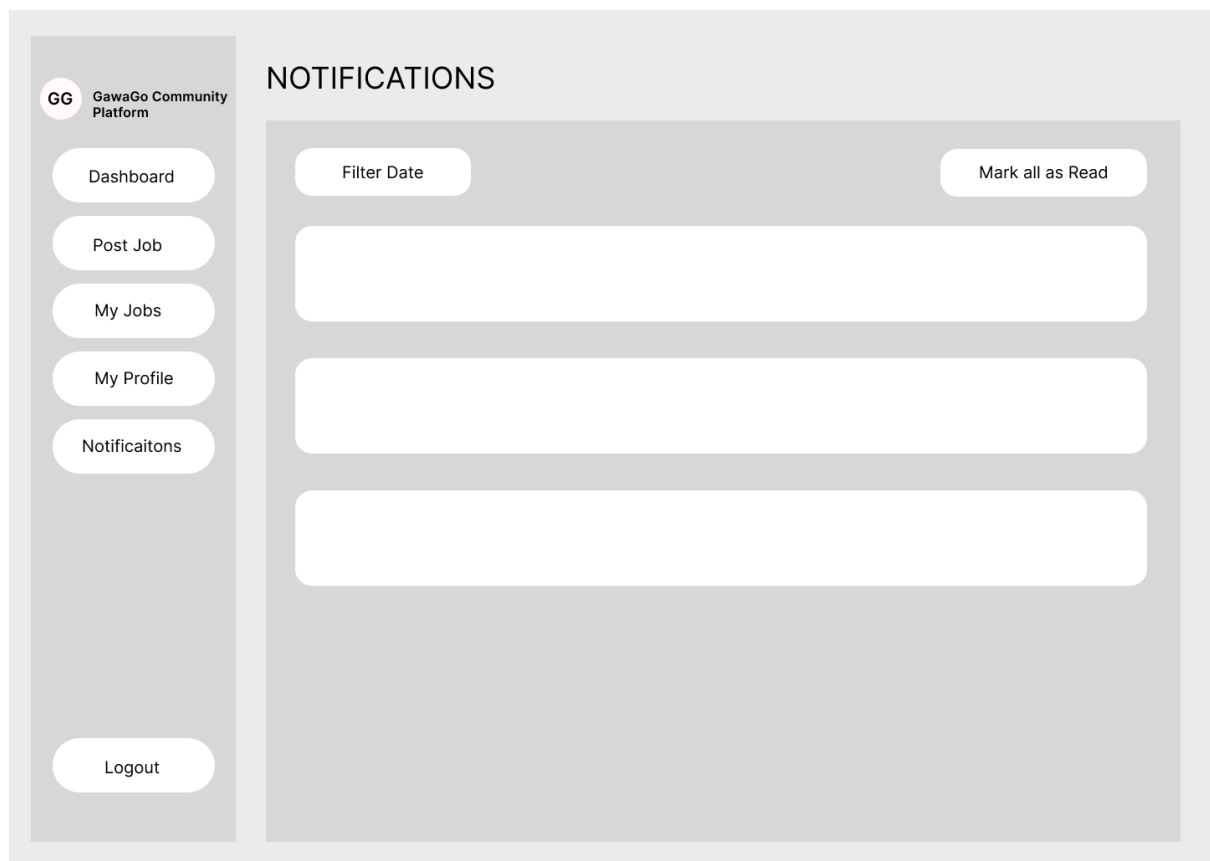


Figure X. Low-Fidelity Prototype of the Notification Interface

The figure presents the low-fidelity prototype of the Notification interface of the GawaGo Community Platform. The design consists of a navigation sidebar and a notification management section where users can view system updates and alerts. Features include options to filter notifications by date and mark all notifications as read. Notification entries are displayed in organized content blocks to ensure clear and efficient communication of important platform activities and updates.

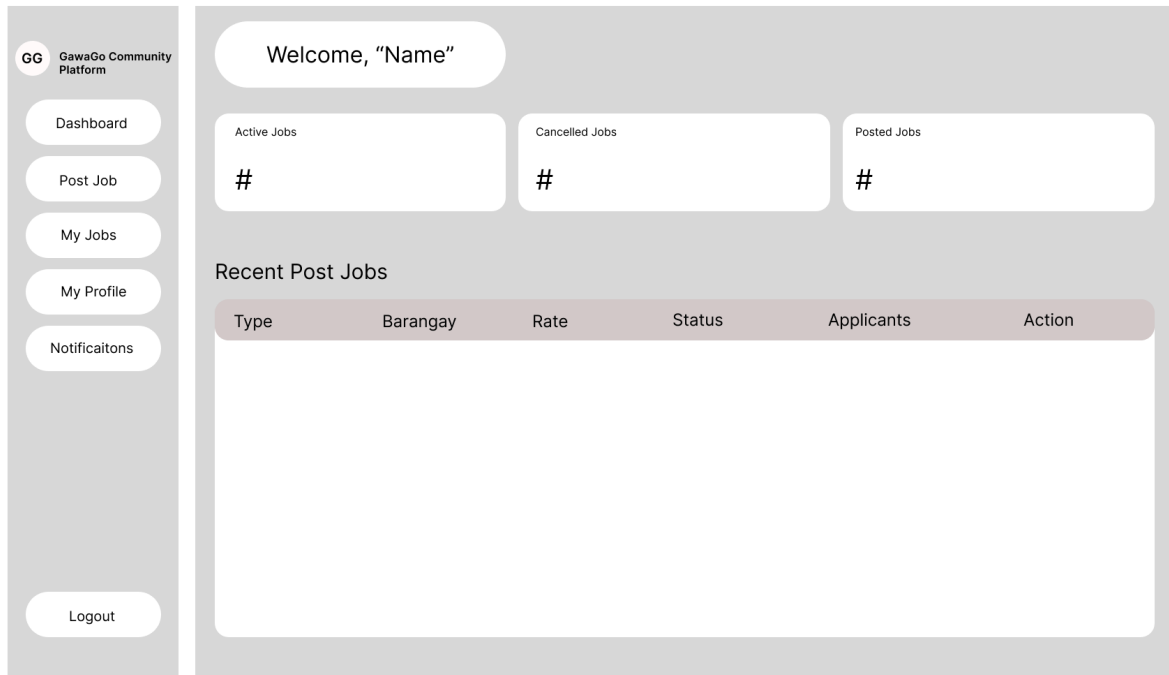


Figure X. Low-Fidelity Prototype of the Household User Dashboard

The figure presents the low-fidelity prototype of the Household User Dashboard in the GawaGo Community Platform. The interface includes a navigation sidebar for accessing key system functions and a dashboard area that provides an overview of job activities, including active, cancelled, and posted jobs. A recent job postings section is also included to allow users to monitor and manage their job requests efficiently. This design provides household users with a centralized view of their job posting activities and system interactions.

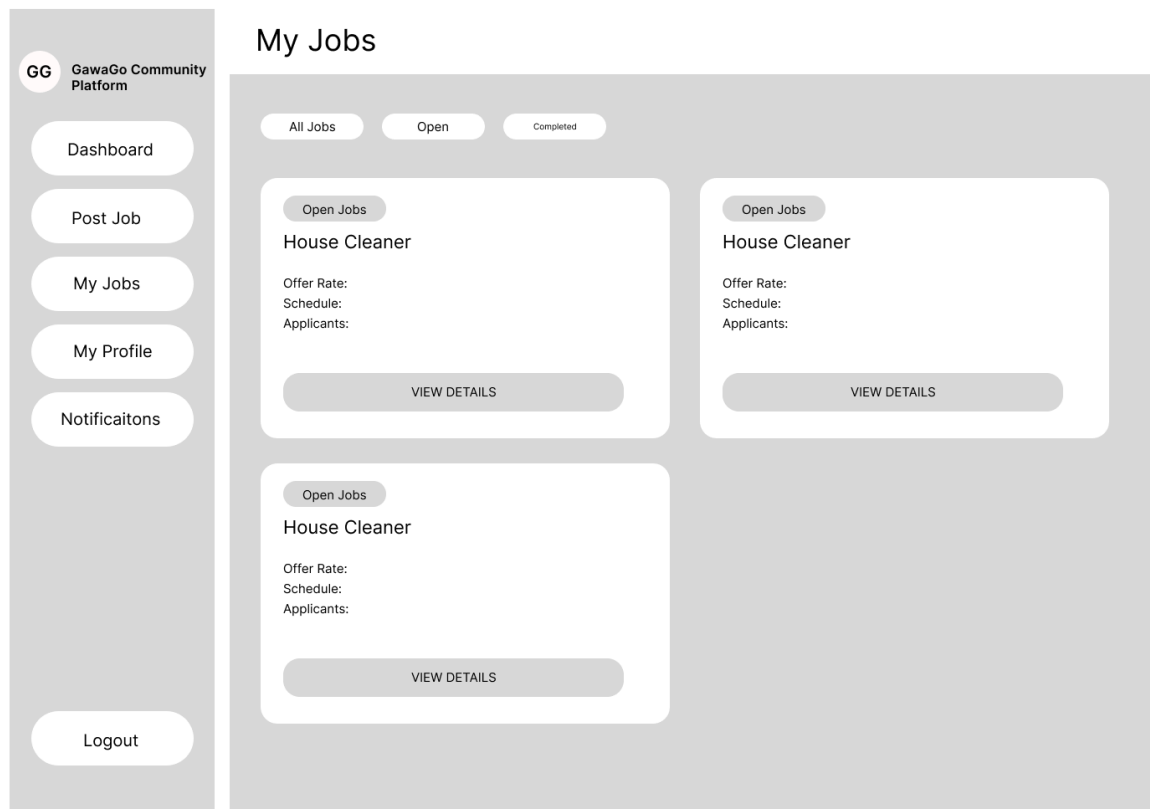


Figure X. Low-Fidelity Prototype of the “My Jobs” Interface

The figure presents the low-fidelity prototype of the “My Jobs” interface of the GawaGo Community Platform. The design includes a navigation sidebar and a job management section that allows users to view and organize their posted jobs. The interface provides tab-based filtering options for all, open, and completed jobs, with a default view of active listings. Job postings are displayed in card formats containing key details such as offered rate, schedule, and number of applicants, along with a view details option for further management. This design supports efficient tracking and organization of user-created job postings.

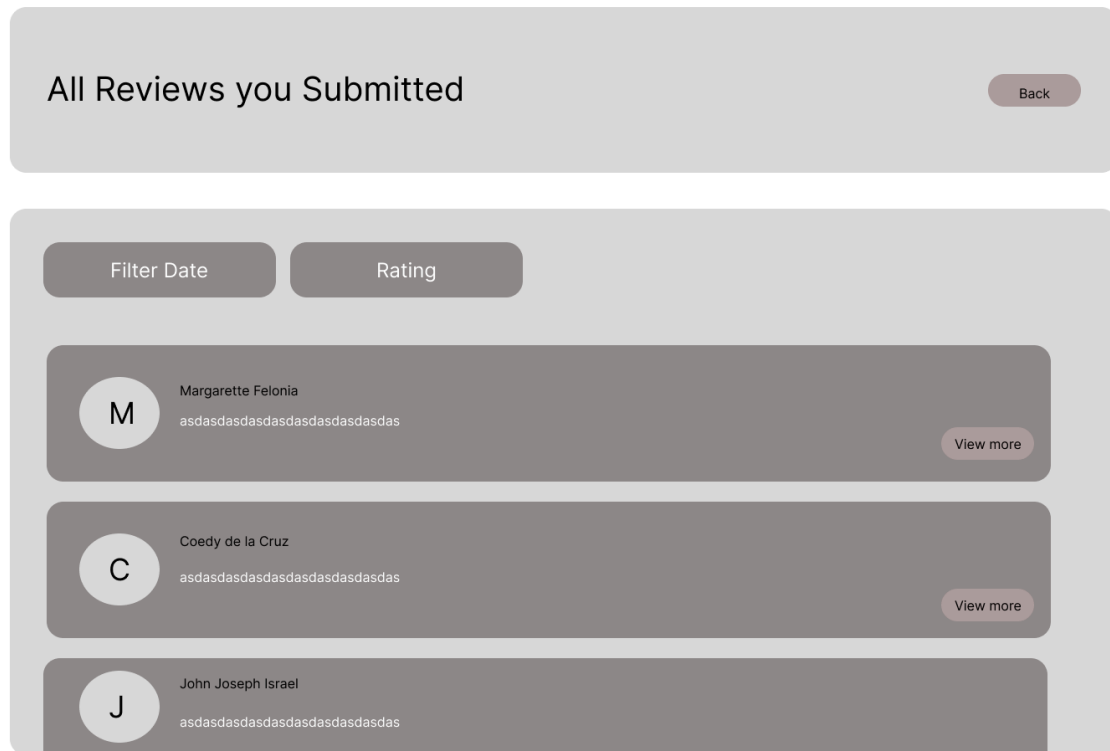


Figure X. Low-Fidelity Prototype of the “All Reviews You Submitted” Interface

The figure presents the low-fidelity prototype of the “All Reviews You Submitted” interface of the GawaGo Community Platform. The design includes a header with navigation and filtering options, allowing users to sort reviews by date and rating. Below the filters, submitted reviews are displayed in structured card layouts containing user identifiers, review content, and an option to view more details. This interface supports organized access and management of user-submitted reviews, enhancing usability and review tracking within the platform.

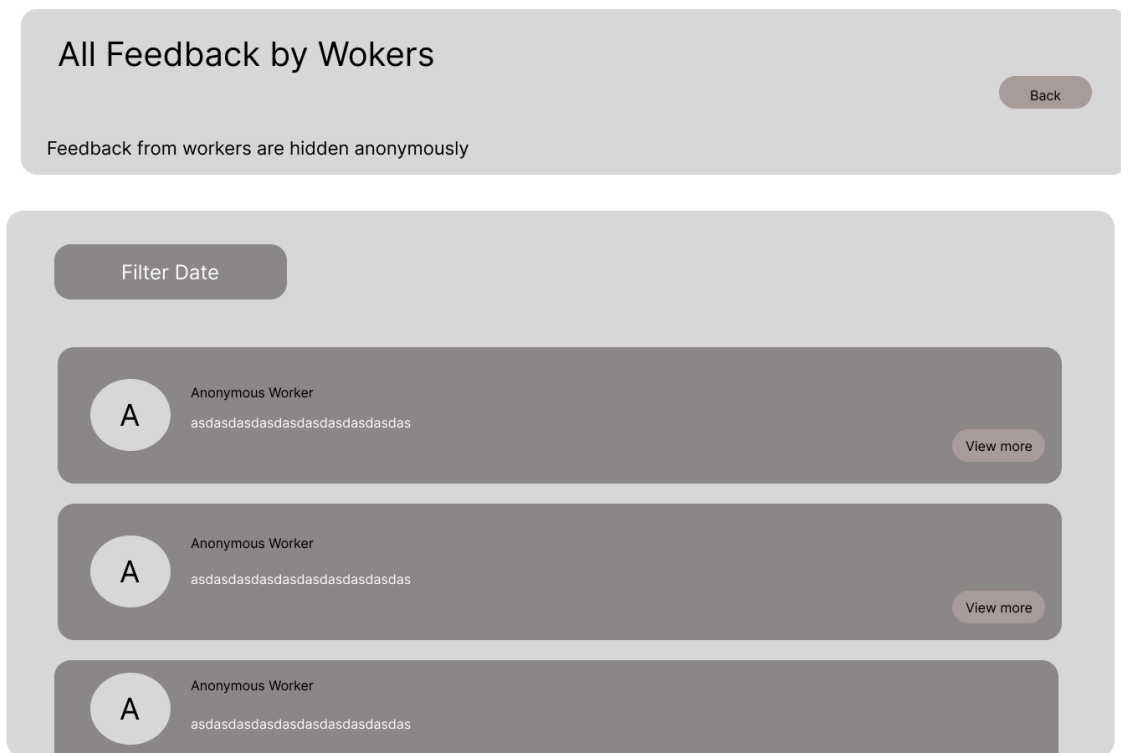


Figure X. Low-Fidelity Prototype of the Worker Feedback Interface

The figure presents the low-fidelity prototype of the Worker Feedback interface of the GawaGo Community Platform. The design includes a header section with navigation controls and a filtering option for sorting feedback by date. The main content area displays anonymized worker feedback in a structured list format, where each entry contains the feedback content and a “View more” option for detailed viewing. This interface supports organized and privacy-focused presentation of worker feedback to assist users in evaluating service quality.

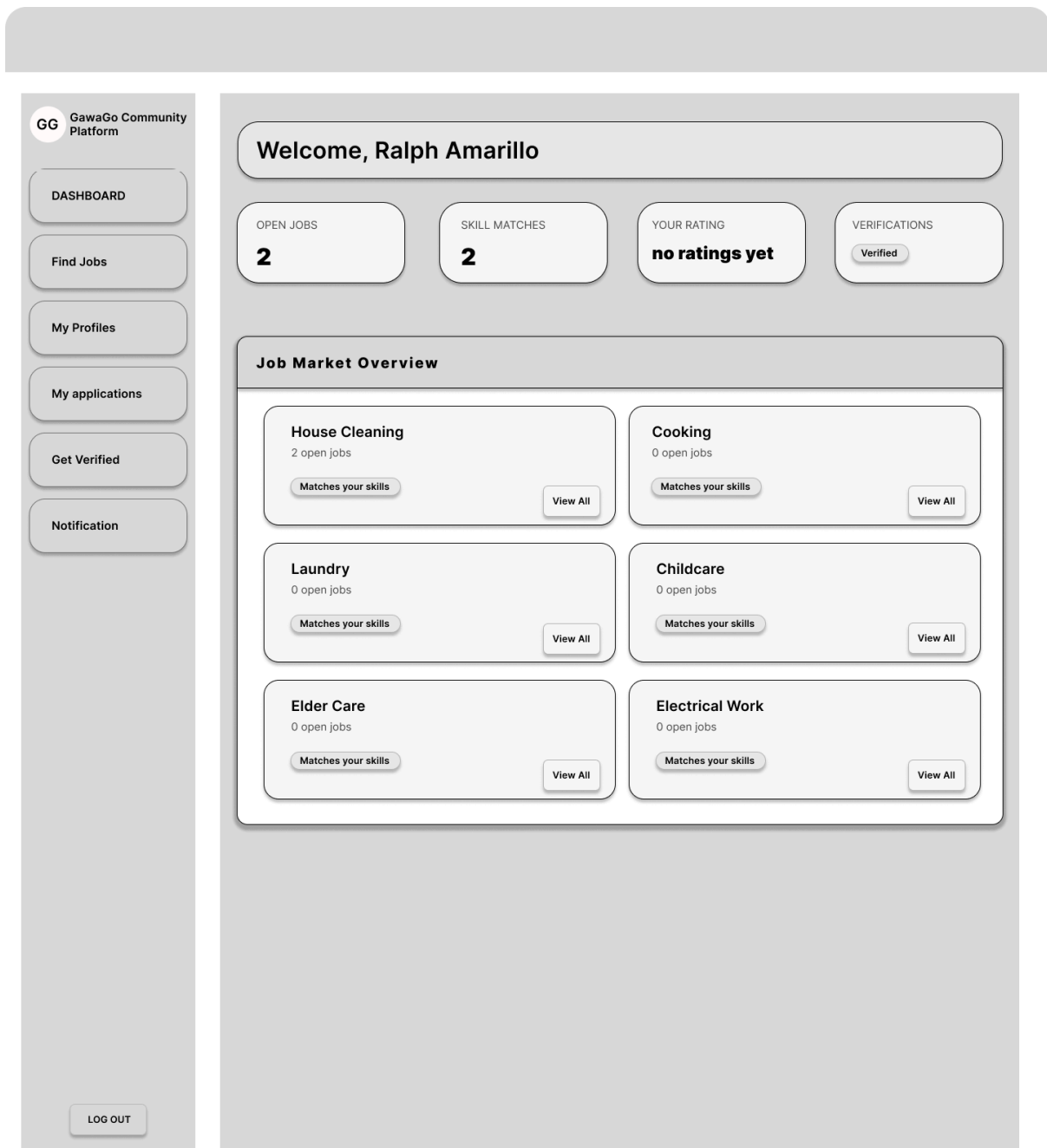


Figure X. Low-Fidelity Prototype of the Worker Dashboard Interface

The figure presents the low-fidelity prototype of the Worker Dashboard interface of the GawaGo Community Platform. The design includes a navigation sidebar and a main dashboard area that provides an overview of user activity and status. Summary cards display key information such as open jobs, skill matches, user rating, and verification status. Below these, a job market overview presents available job categories with indicators for open

positions and skill relevance. This interface supports workers in efficiently monitoring opportunities, matching skills, and managing their platform activities.

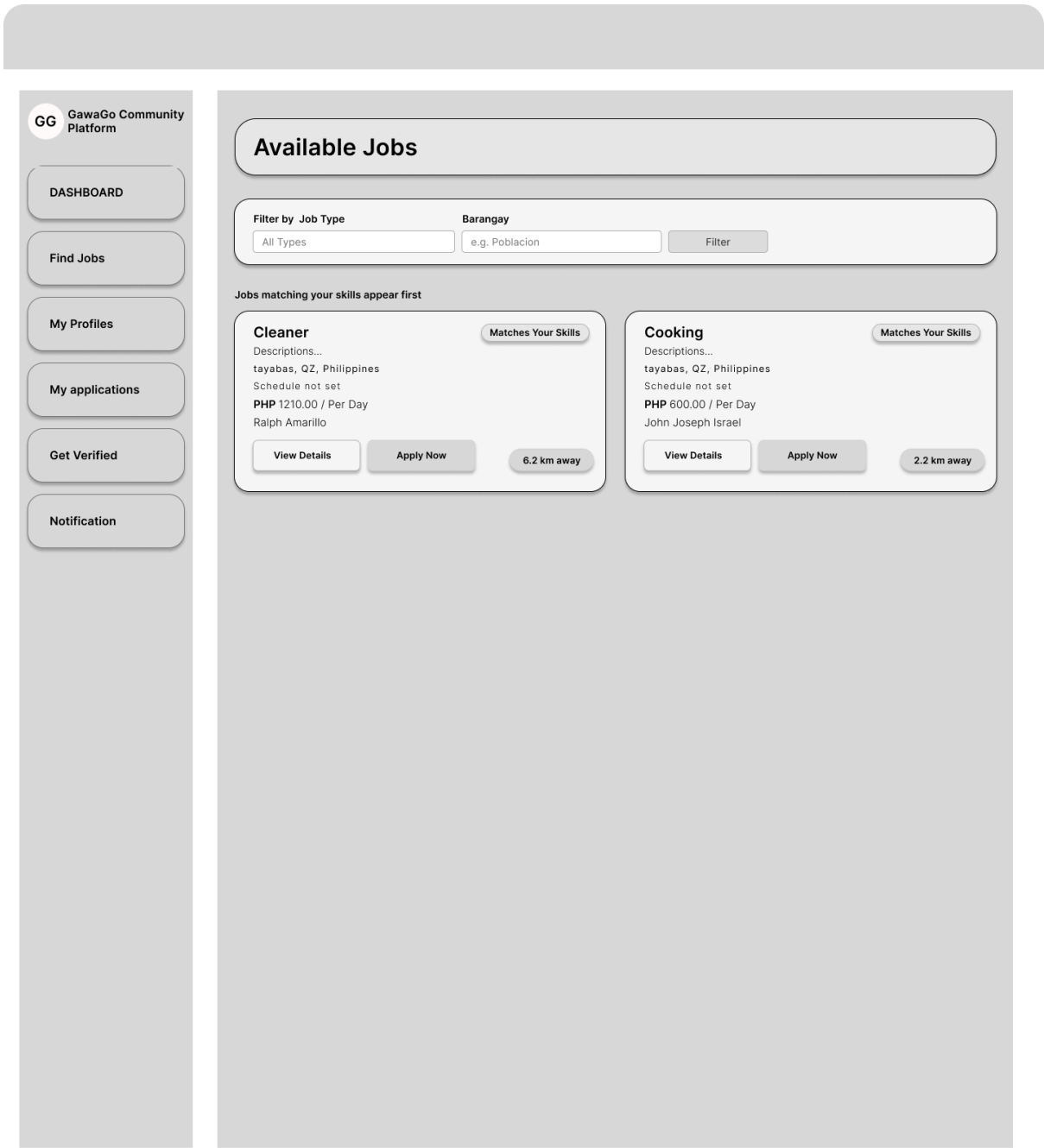


Figure X. Low-Fidelity Prototype of the Available Jobs Interface

The figure presents the low-fidelity prototype of the Available Jobs interface of the GawaGo Community Platform. The design includes a navigation sidebar and a job listing area with filtering options for job type and location. Available job postings are displayed in card

formats containing key details such as role, location, salary, schedule, and distance, along with options to view details and apply. This interface supports efficient job searching and matching based on user skills and preferences.



Figure X. Low-Fidelity Prototype of the Job Details Interface

The figure presents the low-fidelity prototype of the Job Details interface of the GawaGo Community Platform. The design includes a navigation sidebar and a detailed job information view that provides comprehensive information about a selected job posting. The interface displays the household profile, job description, location and distance via map integration, and work area photos for reference. It also includes action buttons for applying to the job or returning to the job listings. This design supports informed decision-making by allowing workers to review complete job details prior to application.

GG

GawaGo Community Platform

DASHBOARD

Find Jobs

My Profiles

My applications

Get Verified

Notification

My Profile

OPEN

RR

Ralph Amarillo

@ralphskiee

Worker

Barangay: Calumpang

Phone: +639233959321

Email: ralphskiee

Verification: Pending

Rating: 5.00

Jobs Done: 3

Hourly Rate: PHP 30.00

Daily Rate: PHP 300.00

Skills: Laundry House Cleaning Childcare Electrical Work Painting Driving

Edit Profile Information

Profile Photo

Choose File

Accepted: JPG, PNG. Clear face photo recommended.

Personal Information

First NameLast Name

EmailPhone Number

BarangayStreet / House No.

Worker Information

Bio / About me

AvailabilityHourly Rate (PHP)Daily Rate (PHP)

Available

Skills (Select all the apply)

☒ House Cleaning

☐ Cooking

☒ Laundry

☒ Childcare

☐ Elder Care

☐ Gardening

☒ Electrical Work

☐ Plumbing

☐ Carpentry

☒ Painting

☐ Aircon Repair / Cleaning

☐ Welding

☒ Driving

☐ Other

Save ChangesCancel

Figure X. Low-Fidelity Prototype of the User Profile and Profile Editing Interface

The figure presents the low-fidelity prototype of the User Profile and Profile Editing interface of the GawaGo Community Platform. The design includes a navigation sidebar and a main content area divided into profile display and profile editing sections. The profile section presents user information such as personal details, location, contact information, verification status, ratings, completed jobs, rates, and skills. The editing section allows users to update personal and worker-related information, including profile details, availability, skills, and

rates. This interface supports comprehensive profile management, enabling users to maintain accurate and updated account information.



Figure X. Low-Fidelity Prototype of the “Get Verified” Interface

The figure presents the low-fidelity prototype of the “Get Verified” interface of the GawaGo Community Platform. The design includes a navigation sidebar and a main content area that displays the user’s verification status. The interface indicates that the user is already verified and presents supporting information such as primary identification, submitted documents, and an admin note confirming approval. This design supports transparency in the

verification process by providing users with a clear view of their verification status and submitted requirements.

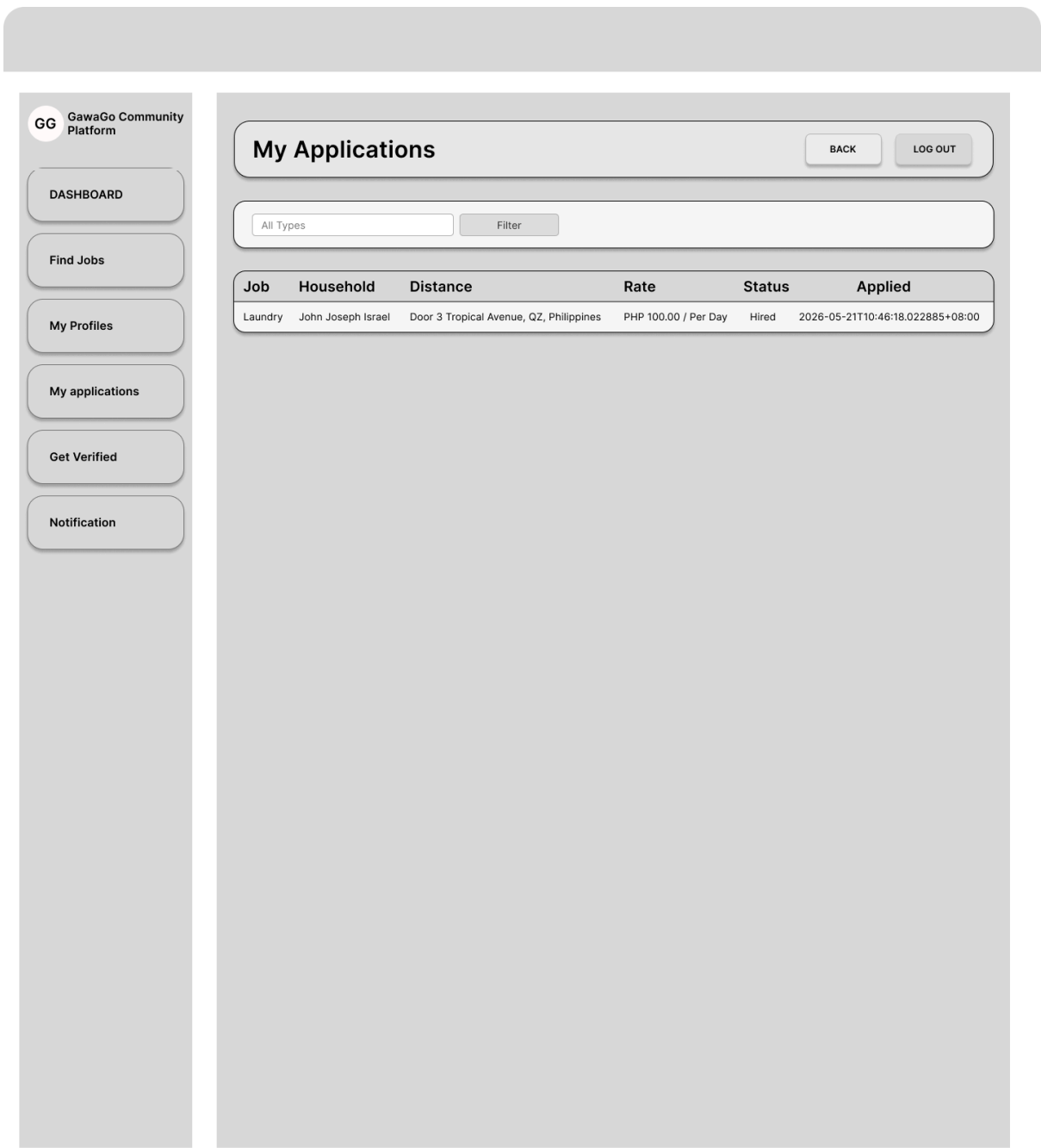


Figure X. Low-Fidelity Prototype of the “My Applications” Interface

The figure presents the low-fidelity prototype of the “My Applications” interface of the GawaGo Community Platform. The design includes a navigation sidebar and a main content area that allows users to view and manage their job applications. The interface provides

filtering options and displays application details in a structured table format, including job information, household name, location, rate, status, and application date. This design supports efficient tracking of application status and history within the platform.

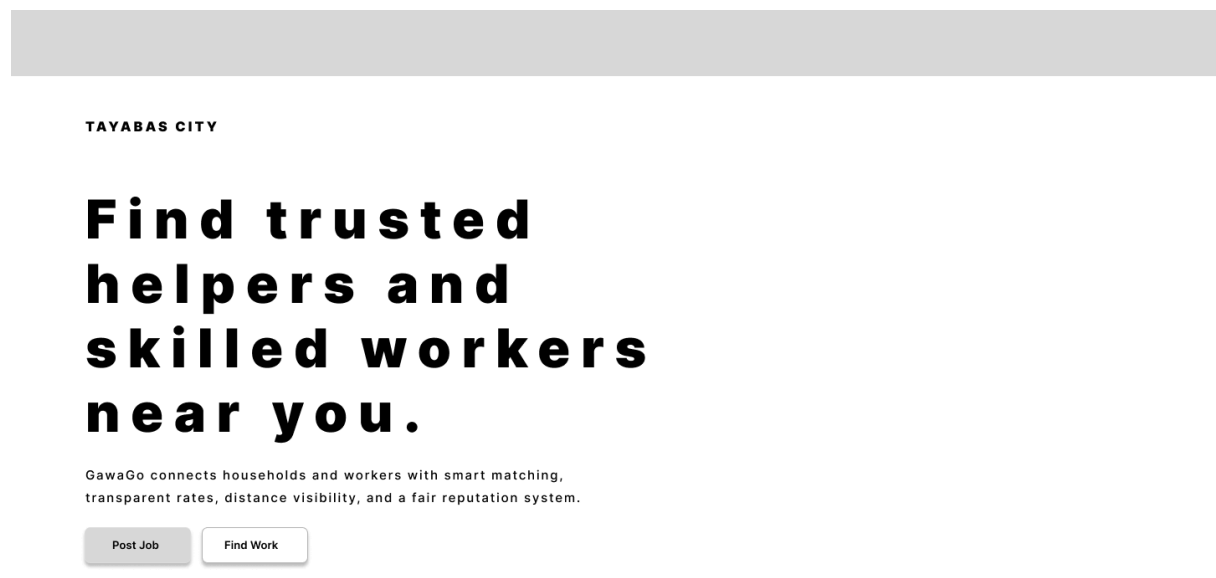


Figure X. Low-Fidelity Prototype of the Landing Page Interface

The figure presents the low-fidelity prototype of the landing page of the GawaGo Community Platform. The design features a minimal layout with a prominent headline introducing the platform's purpose, which is to connect households with skilled workers. Supporting text describes key system features such as smart matching, transparent rates,

distance visibility, and a reputation system. The interface includes primary call-to-action buttons for posting jobs and finding work. This design serves as the entry point of the system, guiding users toward either hiring services or seeking employment opportunities.

Technical Background

Technical Background

This section presents the technical specifications and tools used in the development of the proposed GawaGo Employment Platform. It covers the server-side requirements, client-side requirements, and the development tools employed throughout the system development lifecycle. The technologies described in this section were selected based on their reliability, compatibility with modern web and mobile application development, security features, and overall suitability to the functional requirements of the proposed system.

Server-Side Requirements

The server-side of GawaGo handles the main system functions such as user authentication, database operations, business logic, notifications, and employment analytics. The technologies used were chosen for their reliability, security, and scalability.

- **Operating System** - The development environment uses Windows 11, which provides a stable platform for coding, testing, and debugging. For deployment, the system is intended to run on Ubuntu Server Linux, which is widely used for web hosting due to its stability, security, and performance.
- **Backend Language and Framework** - The system uses Python as the backend language and Django as the framework. Python is chosen for its simplicity and strong support for web development, while Django provides built-in features such as authentication, database management (ORM), and security protection. Its structure supports fast development and easy maintenance for multiple system modules.
- **Database Management System** - During development, the system uses SQLite as the local database through Django. It stores all system data such as user accounts,

profiles, job postings, applications, reviews, notifications, and verification records.

For production, the system may use PostgreSQL for better performance, scalability, and data handling.

- **API Integration Tools** - The system uses APIs mainly for email notifications through Django's email system. It sends alerts for registration, job updates, application status, and verification results. During development, emails are tested through the console, while production will use an SMTP email service. The system currently uses rule-based logic for smart matching and does not rely on external AI APIs.

Client-Side Requirements

The client-side component of the system provides the graphical user interface through which households, workers, and administrators interact with the GawaGo platform. The client-side technologies were selected to ensure a responsive, accessible, and consistent user experience across desktop and mobile devices.

- **Frontend Technologies** - The web-based frontend of the platform is developed using React as the primary JavaScript framework, combined with Bootstrap 5, HTML5, CSS3, and standard JavaScript for interface construction and styling. React enables the development of reusable and modular user interface components that update efficiently without requiring full page reloads, resulting in a smoother and more responsive experience for users. Bootstrap 5 provides a flexible grid-based layout system that ensures the interface adapts properly to different screen sizes and device types, supporting both desktop and mobile users of the web platform. The mobile application component of GawaGo is developed using React Native, which allows the

research team to build a cross-platform mobile client from a shared JavaScript codebase. Both the web and mobile clients connect to the same Django backend API, ensuring that all users receive consistent and synchronized data regardless of the platform they are using to access the system.

- **Browser Compatibility** - The web platform is designed to function correctly on widely used modern browsers. These include Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The system's use of standard HTML5, CSS3, and React-based components ensures consistent behavior and appearance across these browsers. Actual compatibility will be verified during the system testing phase to confirm that all features function as expected across different browser environments.

- **Hardware Minimum Specifications** - The following minimum hardware specifications are recommended for users accessing the GawaGo platform to ensure acceptable system performance. These specifications apply to both desktop and mobile devices used by households, workers, and administrators.

Desktop and Laptop Devices

Component	Specification
Processor	Intel Core i3 or equivalent, running at 1.1 GHz or higher
Memory (RAM)	4 GB minimum
Storage	20 GB available disk space
Internet Connection	Stable broadband connection
Web Browser	Any modern browser (Chrome, Firefox, Edge, or Safari)

Mobile Devices

Component	Specification
Operating System	Android 8.0 (Oreo) or later
Memory (RAM)	3 GB minimum
Internet Connection	Stable mobile data or Wi-Fi connection
Application	Expo Go application or an installed Android APK for testing the mobile app

These specifications are intended as a general guide to ensure that users can interact with the platform without experiencing significant performance issues. Devices that exceed these minimum requirements will generally provide a more responsive and comfortable experience when using GawaGo.

Development Tools

Several tools were used throughout the development lifecycle of the proposed system to support interface design, source code management, and programming assistance. These tools were selected based on their functionality, ease of use, and relevance to the specific tasks involved in the development of GawaGo.

- **Design and Prototyping** - Figma was used as the primary tool for interface design and wireframe development. Figma supports real-time collaborative prototyping and visual design planning, enabling the research team to create, review, and refine both low-fidelity and high-fidelity mockups of the system's web and mobile interfaces

before actual development began. The wireframes and design layouts produced in Figma served as the visual guide for implementing the system's user interface during the development phase.

- **Version Control** - GitHub was used as the primary platform for version control and source code management throughout the development process. GitHub allows the research team to track all changes made to the codebase, manage different versions of the project, collaborate on code contributions, and maintain a complete backup of the system's source files. The use of GitHub also supports documentation and project management activities associated with the development lifecycle.
- **AI-Assisted Development** - The researchers utilized AI-assisted development tools, specifically OpenAI Codex and ChatGPT, during selected stages of the development process. These tools provided support for code generation, identification of programming errors, preparation of technical documentation, and general software development assistance. All outputs generated by these tools were reviewed, validated, and tested by the research team before being incorporated into the system to ensure accuracy, correctness, and alignment with the project's requirements.

Statistical Treatment

Statistical Treatment

The researchers used the Weighted Mean as the primary statistical tool for analyzing and interpreting the evaluation results gathered from the system's respondents. The Weighted Mean provides a single numerical value that represents the overall level of agreement or satisfaction expressed by respondents for each evaluation criterion, making it suitable for summarizing the acceptability of the system based on its measured quality characteristics.

The formula used for computing the Weighted Mean is as follows:

$$WM = \Sigma fx / N$$

Where:

WM = Weighted Mean

Σfx = Sum of the products of each scale value and its corresponding frequency of responses

N = Total number of respondents

The computed Weighted Mean scores for each evaluation criterion were interpreted using the five-point Likert scale presented in Table [X] below. This scale assigns a descriptive rating to each range of computed values to provide a clear and standardized interpretation of the respondents' evaluation results.

Scale	Range	Verbal Interpretation
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5	4.21 – 5.00	Excellent
4	3.41 – 4.20	Very Good
3	2.61 – 3.40	Good
2	1.81 – 2.60	Fair
1	1.00 – 1.80	Poor

Table [X]. Five-Point Likert Scale and Descriptive Rating Interpretation Guide

The resulting weighted mean scores were interpreted according to the scale above to determine the overall quality and acceptability of the proposed GawaGo Employment Platform across all evaluated ISO/IEC 25010 quality characteristics. The complete evaluation results and their corresponding interpretations are presented and discussed in Chapter 4 of this manuscript.

ISO/IEC/IEEE 25010 Evaluation

ISO/IEC 25010 System Evaluation

The quality of the proposed GawaGo Employment Platform was evaluated using the ISO/IEC 25010 Software Product Quality Model as the evaluation framework. ISO/IEC 25010 is an internationally recognized standard for assessing software quality that provides a structured set of quality characteristics and sub-characteristics applicable to software products and systems. Its use in this study ensures that the system is evaluated against objective and widely accepted criteria rather than solely on the subjective impressions of evaluators.

The evaluation was conducted with the participation of selected IT experts, faculty evaluators, and end-users including households, domestic helpers, and skilled workers from Tayabas City who assessed the system after a hands-on demonstration and testing session. The evaluation focused on seven quality characteristics from the ISO/IEC 25010 model that are most relevant to the proposed system's purpose and intended use. These characteristics and their descriptions as applied to the proposed system are presented in the table below.

Quality Characteristic	Description
Functional Suitability	Refers to the degree to which the system provides functions that meet the stated and implied needs of users when operated under the specified conditions of use, including job posting, worker application processing, smart matching, verification management, notification delivery, ratings and reviews, and employment analytics.

Performance Efficiency	Refers to the performance of the system relative to the resources used under specified conditions, encompassing response time, processing speed, dashboard loading performance, and the overall efficiency of database operations during system use.
Compatibility	Refers to the ability of the system to perform its required functions while operating across different devices, web browsers, and operating environments without disrupting existing functionality or negatively affecting user experience.
Usability	Refers to the degree to which the system can be used by its intended users — households, workers, and administrators — to accomplish specific tasks with effectiveness, efficiency, and satisfaction within the context of use.
Reliability	Refers to the degree to which the system consistently performs its required functions without failure over a specified period, including stable operation of core features such as matching, notifications, and data processing.
Security	Refers to the degree to which the system protects user accounts, personal information, verification documents, and platform resources from unauthorized access, data breaches, and other security threats.
Maintainability	Refers to the ease and efficiency with which the system can be modified, corrected, updated, or improved by developers

	throughout its operational lifecycle, including the clarity of code structure and documentation.
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Table [X]. ISO/IEC 25010 Quality Characteristics Used in the System Evaluation

A structured evaluation questionnaire corresponding to these seven quality characteristics was prepared and administered to the evaluation respondents. The responses collected were analyzed using the Weighted Mean formula described in Section 3.8 and interpreted using the five-point Likert scale to determine the overall quality, usability, and acceptability of the GawaGo Employment Platform. The detailed evaluation results are presented and discussed in Chapter 4 of this manuscript.